Who Gets Caught: Blockholder Disclosure Rules and Market Inference

Appendix: statements, hypothesis lists, and proofs

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This appendix carries every proof behind the paper's labelled statements, and closes with two lemmas on the pricing root and the filing timing. It starts with the disclosure partition and the fixed-policy results, then gives the order-size-two pooled experiment, the threshold dial, and the who-gets-caught corollary. The standing conditions (S1) to (S11) apply to every statement that uses them. The corollary's hypotheses (C-1) to (C-3) cite them by number.

1 The disclosure partition and the fixed-policy results

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Lemma 1 (The flagged cell does not move with liquidity)**.** *Fix the cutoff policy and the execution policy, and let the disclosure rule (τ, T) be fixed. Assume the primitive draws are mutually independent with strictly positive variances; the menu, the image of the coarsening Γ, the noise support and the calendar are finite; every stake path $s \mapsto B_j(s, d)$ is Borel, with Voice stakes weakly increasing in the signal and in the date; the blockholder does not revise the stake path in response to realised order flow or prices; the legal-clock definitions hold, including $D = 1 \Rightarrow a = 1$ and the restriction that only Voice plans cross; the composed terminal target $s \mapsto b^*_{j(s)}(s)$ is strictly increasing on the flagged signal region, almost surely on the flagged set; the inner pricing fixed point is unique; the flagged weight satisfies $\Omega > 0$; and the bidder's entry rule has the two properties*

$$\mathbb{P}(\mathsf{B} = 1 \mid \mathcal{I}_H) = p(\mathcal{I}_H) \qquad \textit{and} \qquad \mathsf{B} \perp v \mid \mathcal{I}_H. \tag{1}$$

Then, on the flagged cell $\mathcal{C}_F$, the flagged tuple $\mathsf{S}_F = (B^F, Q^F, a{=}1)$ makes the pre-filing pooled history conditionally independent of (v, s, ξ), and the flagged posterior, the flagged price, the bidder's entry probability and the flagged cell average M_F do not depend on κ. In addition Ω does not depend on κ, so $\partial_\kappa \Omega = 0$ at fixed policies.

Proof. Write $\Xi := (v, s, \xi)$ for the triple whose conditional independence is at issue, $z_{0:H}$ for the noise vector, $\mathcal{H}^P_{f-}$ for the pre-filing pooled history on a flagged path, and $\mathcal{I}_H$ for the public information set at the control node, which the model sets equal to S_F on the flagged cell. The functions u_1, u_2, u_3, u_5 below are bounded and measurable and are local to this proof.

Step 1 (where κ lives). At fixed cutoff and execution policies, and with no revision of the path within the window, every policy object is a function of $(s;\tau,T)$ alone: the selected plan $j=j(s)$, the stake path $B_j(s,\cdot)$, the order marks $q_{jd}(s)$, the terminal target $b_j^*(s)$, the crossing date $c_j(s)$, the filing date $f_j(s)$, the disclosure indicator $D_j(s)$, the stake at filing $B_j^F(s)$ and the flagged order $Q_j^F(s)$. Absence of feedback removes any dependence on realised order flow or prices, and fixed policies remove any dependence on κ through re-optimised cutoffs. Among the primitives, κ appears in exactly one place, the law of the ternary noise mark; the laws of v, ε and ξ and the remaining constants carry no κ. Write Q_κ for the law of $z_{0:H}$. At fixed policies, then, κ enters the model only through Q_κ.

Step 2 (the flagged set and its mass are free of κ). The disclosure indicator $D=\mathbf{1}\{a=1,\ c(\tau)+T\le H\}$ is measurable, and by Step 1 it is a function of the signal alone. Hence the flagged cell $\mathcal{C}_F=\{D=1\}$ lies in $\sigma(s)$ and is one fixed Borel subset of the signal line, the same subset at every κ. The signal $s=v+\varepsilon$ is $N(\mu_v,\sigma_v^2+\sigma_\varepsilon^2)$, a law free of κ because the noise mark enters neither v nor ε. Therefore $\Omega=\mathbb{P}(s\in\mathcal{C}_F)$ is the same number at every κ, which is the last conclusion of the statement.

Step 3 (the flagged tuple pins the signal, with a measurable inverse). On $\mathcal{C}_F$ the engagement flag is $a=1$, since disclosure implies engagement and only Voice plans cross, so the third coordinate of S_F is uninformative there. By the definition $Q_j^F=b_j^*(s)-B_j^F$,

$$B_{j(s)}^F(s)\ +\ Q_{j(s)}^F(s)\ =\ b_{j(s)}^*(s). \tag{2}$$

The composed terminal target on the right of (2) is strictly increasing in the signal on the flagged region and is therefore injective there. That monotonicity is consumed almost surely on the flagged set, which is the only coherent reading when B^F is continuum-valued: no individual flagged tuple then carries positive probability, so injectivity holds almost surely on a set of positive probability rather than on a set of atoms. Consequently the first two coordinates of S_F determine $b_{j(s)}^*(s)$, which determines s, which at fixed policies determines $j=j(s)$. Write ι_F for the resulting inverse. It is strictly increasing on the image of the composed target, and a monotone real function on a Borel set is Borel, so ι_F is measurable; no measurable-selection theorem is needed. Since $\Omega>0$, the set on which this holds carries strictly positive probability and the statement is not vacuous. Restricted to $\mathcal{C}_F$, therefore, $\sigma(\mathsf{S}_F)=\sigma(s)$ up to null sets. Strict monotonicity is what the step consumes: if two signals produced the same flagged tuple while carrying different pooled paths, (2) would fail to be injective and the tuple would not pin the pooled path.

Step 4 (the pre-filing pooled history is a noise transform indexed by plan and signal). Define, for a plan j and a signal s,

$$\Upsilon_{j,s}(z_{0:H})\ :=\ \Big(\big(q_{j0}(s)+z_0,\ldots,q_{j,f_j(s)-1}(s)+z_{f_j(s)-1}\big);\ \text{flag not landed by } f_j(s)-1\Big), \tag{3}$$

so that $\mathcal{H}_{f^-}^P=\Upsilon_{j(s),s}(z_{0:H})$ pointwise on $\mathcal{C}_F$. The indexing by (j,s) alone is exactly the absence of feedback: with re-optimisation inside the window the marks would depend on realised flow, hence on past noise, and no such indexing would exist. The map (3) is jointly measurable in $(s,z_{0:H})$. The filing date takes finitely many values because the calendar is finite, and on each of its finitely many measurable level sets the map is $(s,z)\mapsto(q_{j(s)d}(s)+z_d)_{d<t}$, measurable because $q_{jd}=\Gamma\big(B_j(\cdot,d)-B_j(\cdot,d-1)\big)$ composes a Borel function, Voice stakes being monotone in the

signal and only Voice plans reaching $\mathcal{C}_F$, with the coarsening Γ, itself Borel because it is ordered in the increment. This is unaffected by the size of the blockholder's per-round order, which fixes the value the mark takes on a building round but not its measurability in the signal. Patching finitely many measurable restrictions gives a measurable map. The second component of (3) is a deterministic function of s by Step 1 and carries no information beyond s.

Step 5 (observed prices enlarge nothing). Each pooled price is the unique fixed point of the pricing map at its pooled history and is therefore a function of that history, and likewise the flagged price is a function of the flagged information set. Adjoining the value of a function of a generator of a σ-algebra to that σ-algebra does not enlarge it, so prices may be ignored when the information content of $\mathcal{I}_H$ is computed.

Step 6 (an information sandwich). On $\mathcal{C}_F$,

$$\sigma(s) \cap \mathcal{C}_F \;\subseteq\; \mathcal{I}_H \;\subseteq\; \sigma(s, z_{0:H}) \cap \mathcal{C}_F. \tag{4}$$

For the left inclusion: the filing reveals $F = (B^F, a{=}1)$ and the flagged order is submitted and priced, so S_F belongs to $\mathcal{I}_H$; by Step 3 the tuple recovers s; and $\mathcal{C}_F \in \sigma(s)$ by Step 2, so restricting to the flagged cell is itself an operation within $\sigma(s)$. For the right inclusion: every component of $\mathcal{I}_H$, namely pooled order flow, flag status, filing content, the flagged order and prices, is a measurable function of $(s, z_{0:H})$ by Steps 1, 4 and 5. Everything below uses only (4), so the conclusion is robust to the exact content assigned to the control-node information set, provided it satisfies the sandwich.

Step 7 (a freezing lemma). Let $\mathcal{G}$ be a σ-algebra with $z_{0:H}$ independent of $\mathcal{G}$, let S be $\mathcal{G}$-measurable, let w be bounded and jointly measurable, and put $\bar{w}(x) := \int w(x,\zeta)\, Q_\kappa(\mathrm{d}\zeta)$. Then $\mathbb{E}[w(S, z_{0:H}) \mid \mathcal{G}] = \bar{w}(S)$. For a product form $w(x,\zeta) = u_1(x)u_2(\zeta)$ this is $\mathbb{E}[u_1(S)u_2(z_{0:H}) \mid \mathcal{G}] = u_1(S)\mathbb{E}[u_2(z_{0:H})] = \bar{w}(S)$, pulling out the $\mathcal{G}$-measurable factor and using independence. The product forms generate the product σ-algebra and are closed under multiplication; both sides of the asserted equality are linear in w and stable under bounded monotone limits, so the monotone class theorem extends the equality to every bounded jointly measurable w.

Step 8 (the conditional independence). By Step 3, conditioning on $\sigma(\mathsf{S}_F)$ on the flagged set is conditioning on $\sigma(s)$, so what has to be shown is $\mathcal{H}^P_{f^-} \perp \Xi \mid s$ on $\mathcal{C}_F$. Given s, the middle coordinate of Ξ is degenerate, so it suffices to show, for bounded measurable u_1 on the pre-filing history space and bounded measurable u_2 on $\mathbb{R}^2$,

$$\mathbb{E}\big[u_1(\mathcal{H}^P_{f^-})\, u_2(v,\xi) \mid s\big] \;=\; \mathbb{E}\big[u_1(\mathcal{H}^P_{f^-}) \mid s\big] \cdot \mathbb{E}\big[u_2(v,\xi) \mid s\big] \qquad \text{a.s. on } \mathcal{C}_F, \tag{5}$$

which is the standard characterisation of conditional independence given a σ-algebra. Put $w(x,\zeta) := u_1\big(\Upsilon_{j(x),x}(\zeta)\big)$, bounded and jointly measurable by Step 4, so that $u_1(\mathcal{H}^P_{f^-}) = w(s, z_{0:H})$. The noise vector is independent of (v, ε, ξ) and hence of Ξ, which is a measurable function of that triple. Apply Step 7 with $\mathcal{G} = \sigma(v, s, \xi)$ and $S = s$: $\mathbb{E}[w(s, z_{0:H})u_2(v,\xi) \mid v, s, \xi] = u_2(v,\xi)\, \bar{w}(s)$. Taking $\mathbb{E}[\,\cdot \mid s]$ of both sides and pulling out the $\sigma(s)$-measurable factor $\bar{w}(s)$ gives $\mathbb{E}[u_1 u_2 \mid s] = \bar{w}(s)\, \mathbb{E}[u_2 \mid s]$; setting $u_2 \equiv 1$ in the same display gives $\mathbb{E}[u_1 \mid s] = \bar{w}(s)$, and substituting yields (5). Unconditionally the pre-filing pooled history depends strongly on the signal through the order marks, and the conditioning event removes that dependence by pinning the signal, leaving the noise as the only remaining randomness in (3).

Step 9 (the flagged posterior). Let u_3 be bounded, $\mathcal{I}_H$-measurable and supported on $\mathcal{C}_F$.

By the right inclusion of (4) and the Doob–Dynkin lemma, $u_3 = u_5(s, z_{0:H})$ for some bounded jointly measurable u_5. Running the computation of Step 8 with u_5 in place of w gives $\mathbb{E}[u_2 u_3] = \mathbb{E}\big[\bar{u}_5(s)\,\mathbb{E}[u_2 \mid s]\big]$, while the tower property together with $\mathbb{E}[u_3 \mid s] = \bar{u}_5(s)$ from Step 7 gives $\mathbb{E}\big[u_3\,\mathbb{E}[u_2 \mid s]\big] = \mathbb{E}\big[\bar{u}_5(s)\,\mathbb{E}[u_2 \mid s]\big]$. Hence $\mathbb{E}[u_2 u_3] = \mathbb{E}\big[\mathbb{E}[u_2 \mid s]\,u_3\big]$ for every such u_3; since $\mathbb{E}[u_2 \mid s]$ is $\sigma(s)$-measurable and the left inclusion of (4) places $\sigma(s) \cap \mathcal{C}_F$ inside $\mathcal{I}_H$, it is a version of $\mathbb{E}[u_2 \mid \mathcal{I}_H]$ on the flagged set:

$$\mathbb{E}[u_2(v,\xi) \mid \mathcal{I}_H] \;=\; \mathbb{E}[u_2(v,\xi) \mid s] \qquad \text{a.s. on } \mathcal{C}_F. \tag{6}$$

The extension from bounded to integrable u_2, needed at Step 11 for $u_2(v,\xi) = v$, follows by applying (6) to the truncations $\max(-n, \min(n, u_2))$ and passing to the limit under conditional dominated convergence, the dominating variable being $|u_2|$, integrable because v is Gaussian. Separately, the engagement posterior satisfies $\pi(\mathcal{I}_H) = 1$ on $\mathcal{C}_F$, because the flagged cell lies inside $\{a = 1\}$, and because the cell is itself $\mathcal{I}_H$-measurable, being $\{D = 1\}$ with D part of the flag status carried in $\mathcal{I}_H$.

Step 10 (the posterior does not move with κ). By (6) the conditional law of (v, ξ) given $\mathcal{I}_H$ on the flagged set equals its conditional law given s. That law is $v \mid s \sim N\big(\mu_v + \beta(s - \mu_v), (1-\beta)\sigma_v^2\big)$ with $\beta = \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2)$, and $\xi \mid s \sim N(0, \sigma_\xi^2)$ independent of v. Neither depends on κ, which by Step 1 lives only in Q_κ and enters no coordinate of Ξ. With $\pi = 1$ from Step 9, the flagged posterior over (v, a, ξ) is a function of the signal that does not move with κ, hence by Step 3 a function of S_F that does not move with κ.

Step 11 (the flagged price). The flagged price solves the inner fixed point $P = \mathbb{E}[Y \mid \mathcal{I}_H]$. On the flagged cell $a = 1$ and $\pi = 1$, so $m_0 + \Delta_m = m_1$. Only the two entry properties (1) are used. The entry rule of the model has both, since the bidder's shock satisfies $\xi \mid \mathcal{I}_H \sim N(0, \sigma_\xi^2)$ independently of v by Step 10 and the entry indicator is a function of ξ and of $\mathcal{I}_H$-measurable quantities; any other rule with these two properties works as well. Under them,

$$P \;=\; \big(1 - p(P)\big)\Big(\mu_v + \beta(s - \mu_v) + \Delta_V\Big) \;+\; p(P)\big(P + m_1\big), \qquad p(P) \;=\; 1 - \Phi\!\left(\frac{P + K + m_1 - \bar{S}}{\sigma_\xi}\right), \tag{7}$$

where $\mathbb{E}[v \mid \mathcal{I}_H] = \mathbb{E}[v \mid s] = \mu_v + \beta(s - \mu_v)$ by (6). The data of (7) are the single number $\mathbb{E}[v \mid s]$ and the primitive parameters, none of which depends on κ by Step 10. Uniqueness of the inner fixed point makes the solution a function of the data alone, so equal data give equal solutions. Uniqueness is load-bearing rather than decorative: without it an extraneous selection among fixed points could vary with κ even though the map does not. Hence the flagged price is a function of the signal, and by Step 3 of S_F, that does not move with κ.

Step 12 (the entry probability). With $\pi = 1$ from Step 9, the flagged entry probability is $p = 1 - \Phi\big((P^F + K + m_1 - \bar{S})/\sigma_\xi\big)$ on the flagged cell. Every argument is free of κ, by Step 11 for the price and by the primitive table for the constants. Write $p_F(s)$ for it. It is measurable as a composition of the monotone map $s \mapsto \mathbb{E}[v \mid s]$, the fixed-point solution and the normal distribution function, and it takes values in $(0, 1)$.

Step 13 (the flagged cell average). The kernel of the premium is $h = \pi p$, so $h = p_F(s)$ on the flagged cell by Steps 9 and 12. Since $\Omega > 0$, the elementary definition of conditioning on a positive-probability event gives

$$M_F \;=\; \Delta_m\,\mathbb{E}[h \mid D=1] \;=\; \frac{\Delta_m\,\mathbb{E}\big[p_F(s)\,\mathbf{1}\{s\in\mathcal{C}_F\}\big]}{\Omega}. \tag{8}$$

The numerator of (8) integrates a bounded measurable function free of κ (Step 12) over a Borel set free of κ (Step 2) against the law of the signal, itself free of κ (Step 2), and the denominator is free of κ and strictly positive. Hence M_F does not move with κ, which completes the four invariance conclusions. $\square$

Two boundaries of the result are part of it. Nothing above asserts that the pre-filing pooled price does not move with κ: that price conditions on order flow whose law is Q_κ, and it moves with κ in general. With $J = P^F - P_{\mathrm{ND}}$, where P_{ND} is the last pre-filing pooled price at the same realised order flow, and with $\partial_\kappa P^F = 0$ on the flagged set, wherever the derivative exists $\partial_\kappa J = -\partial_\kappa P^P_{f-}$. The result therefore makes no claim about the filing-day jump. And the result is a fixed-policy statement: if cutoffs or execution paths move with κ, Step 1 is unavailable and the conclusions do not follow.

1.1 The partition and the factorisation

The disclosure rule (τ, T) acts on the model first as a partition of histories and only then as a statement about prices. This subsection separates the two. The partition is combinatorial and needs no restriction on cell mass; the two-cell decomposition of the premium needs an interior partition; the factorisation of the liquidity sensitivity needs, in addition, that the flagged endpoint and the flagged weight are both free of κ.

Standing conditions

The following are in force throughout the subsection.

(S1) *One probability space.* The primitive vector, the value v, the signal noise ε, the bidder draw ξ and the noise-trader marks $z_{0:H}$, has a joint law on a finite product of Polish spaces, with the noise marks valued in the finite set $\{-\bar z, 0, +\bar z\}$. The signal is $s = v + \varepsilon$.

(S2) *A finite menu and a finite calendar.* The plan menu $\mathcal{J}$ is finite, the horizon H is finite, and the window margin satisfies $T \in \{1, \dots, H\}$.

(S3) *A cutoff selection map.* The signal is carried into a plan by a step function $j(\cdot)$ with breakpoints $k_1 \le \cdots \le k_{J-1}$.

(S4) *Monotone Voice paths and a clean start.* Each plan j carries a stake path $B_j(s, \cdot)$ with $B_j(s, -1) = b_0 < \tau$; for a Voice plan the path is weakly increasing in the signal and weakly increasing in the calendar date.

(S5) *Legal-clock discipline.* Only Voice plans cross the threshold; the crossing date $c_j(s)$ is the first date at which the path reaches τ, or $+\infty$ if the path never reaches it; the filing lands exactly at $f_j = c_j + T$ and only through the disclosure node; the filing reports the stake truthfully.

(S6) *The flag is public.* Each pooled history carries the coordinate "the filing has landed by date d", and the control-node information set $\mathcal{I}_H$ contains the pooled history up to the control node.

(S7) *No-feedback timing.* The executed path, the order marks, the terminal target, the crossing date, the filing date, the filing stake and the flagged order are functions of the plan and the signal alone. Neither realised order flow nor a realised price enters any of them.

(S8) *A bounded, pinned kernel.* The engagement-premium kernel is $h(\mathcal{I}) = \pi(\mathcal{I})\, p(\mathcal{I})$, where the engagement posterior $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I})$ takes values in $[0, 1]$ and the bidder-entry probability $p(\mathcal{I})$ takes values in $(0, 1)$ and is a continuous function of the posterior and the price. The pricing rule pins one version of $\mathbb{E}[Y \mid \mathcal{I}]$ rather than an equivalence class.

(S9) *A finite wedge.* The premium wedge Δ_m is a finite, strictly positive constant, and $\Delta_{\text{act}} = \Delta_m\, \mathbb{E}[h(\mathcal{I}_H)]$.

(S10) *Liquidity enters in one place.* The intensity κ appears in the primitives only in the law of the ternary noise mark. The laws of v, ε and ξ and the remaining constants carry no κ.

(S11) *Fixed policies.* The plan menu, the execution policies and the cutoff vector k are held fixed in κ and held fixed across any two rules compared.

The partition

Label: PROVED.

Lemma 2 (Disclosure partition)**.** *Under (S1)–(S6), the disclosure indicator $D = \mathbf{1}\{a = 1,\ c(\tau) + T \le H\}$ is a measurable function of the control-node history, and the flagged cell $\mathcal{C}_F = \{D = 1\}$ and the pooled cell $\mathcal{C}_P = \{D = 0\}$ are disjoint and cover the space. Consequently the flagged weight $\Omega = \mathbb{P}(D = 1)$ is defined and $\mathbb{P}(D = 0) = 1 - \Omega$. No restriction on the mass of either cell is used.*

Proof. Step 1 (the selection map is Borel). By (S1) the signal $s = v + \varepsilon$ is a sum of two coordinates of a Borel vector and is therefore Borel. By (S3) the map $j(\cdot)$ is a step function whose breakpoints are finite in number, by (S2), so it is a finite sum of indicators of half-lines and hence Borel. The set $\{s : j(s) = j\}$ is Borel for each $j \in \mathcal{J}$.

Step 2 (the engagement flag is measurable). The realised flag is $a = a_{j(s)}$, the composition of the Borel map of Step 1 with the map $j \mapsto a_j$ defined on the finite set $\mathcal{J}$. A map on a finite set carrying its discrete σ-algebra is measurable, so $\{a = 1\}$ is Borel and lies in $\sigma(s)$.

Step 3 (the crossing date is attained, unique and measurable). Fix a Voice plan j. By (S4) the map $s \mapsto B_j(s, d)$ is weakly increasing, and the preimage of $[\tau, \infty)$ under a weakly monotone real function is an interval, so $\{s : B_j(s, d) \ge \tau\}$ is Borel at each date d. The calendar $\{0, \dots, H\}$ is finite by (S2), so

$$\{c_j \le d\} \;=\; \bigcup_{d' \le d} \{s : B_j(s, d') \ge \tau\}$$

is a finite union of Borel sets and c_j is a Borel map into $\{0, \dots, H\} \cup \{+\infty\}$. Its value is unique: a nonempty subset of the finite well-ordered set $\{0, \dots, H\}$ has exactly one minimum, and by (S5) the crossing date is that minimum; if the subset is empty then $c_j = +\infty$.

Step 4 (the disclosure indicator is measurable). By (S5) a filing requires $a = 1$, and $f_j = c_j + T \le H$ holds exactly when $c_j \le H - T$, the convention $(+\infty) + T = +\infty > H$ being available because H is finite by (S2). Hence

$$\{D = 1\} \;=\; \bigcup_{j \in \mathcal{J} \,:\, a_j = 1} \Big(\{s : j(s) = j\} \;\cap\; \{s : c_j(s) \le H - T\} \Big). \tag{9}$$

Each set on the right is Borel by Steps 1 and 3, and the union is finite by (S2), so $\{D = 1\}$ is Borel and D is a measurable $\{0,1\}$-valued map. Record one fact for later: at a fixed cutoff policy every ingredient of (9) is a function of s alone, so $\{D = 1\} \in \sigma(s)$. Record a second: regularity of $s \mapsto B_j(s, d)$ was needed only for Voice plans, the conjunct $a = 1$ having removed the others from the union.

Step 5 (exactly one cell, at the level of states). By Step 4, D is the indicator of a Borel event, so at every sample point it takes exactly one of the values 0 and 1. Therefore $\mathcal{C}_F \cap \mathcal{C}_P = \varnothing$ and $\mathcal{C}_F \cup \mathcal{C}_P$ is the whole space.

Step 6 (the cell is a function of the observed history). Step 5 partitions the space of states; the claim is about control-node histories. By (S6) each pooled history carries the coordinate "the filing has landed by date d", so the flag is public by construction, and $\mathcal{I}_H$ contains that coordinate at $d = H$. By (S5) the filing lands exactly at $c + T$ and only through the disclosure node, so the coordinate "the filing has landed by H" equals $\mathbf{1}\{f \le H\}$; combining this with $D = \mathbf{1}\{a = 1,\ c < \infty,\ f \le H\}$ and with the restriction that only Voice plans cross, that coordinate equals D. Hence D is measurable with respect to the control-node history and the map from a history to its cell is single-valued. When both cells have positive probability, the map is onto $\{\mathcal{C}_F, \mathcal{C}_P\}$. When $\Omega \in \{0, 1\}$, one cell is null and the map hits only the other almost surely, as Step 7 records. The public flag coordinate is what carries this step: without it D would still be a measurable function of the state by Step 4, while the cell assignment would stop being a function of what is observed at the control node.

Step 7 (the weight). By Steps 5 and 6 the two cells are measurable, disjoint and exhaustive, so $\Omega = \mathbb{P}(D = 1)$ is defined and $\mathbb{P}(D = 0) = 1 - \Omega$. Steps 1 to 7 used no restriction on the value of Ω, which may be 0 or 1. □

The two-cell decomposition

Label: PROVED.

Lemma 3 (Two-cell decomposition of the premium)**.** *Under (S1)–(S9), with the partition of Lemma 2, and whenever* $0 < \Omega < 1$,

$$\Delta_{\text{act}} \;=\; \Omega\, M_F \;+\; (1 - \Omega)\, M_P, \qquad M_F = \Delta_m \mathbb{E}[h \mid D = 1], \quad M_P = \Delta_m \mathbb{E}[h \mid D = 0]. \tag{10}$$

At $\Omega = 1$ *the identity reads* $\Delta_{\text{act}} = M_F$ *and at* $\Omega = 0$ *it reads* $\Delta_{\text{act}} = M_P$*; in each of those two cases the average over the null cell is undefined rather than zero, and it is not determined by the law. That* M_P *is defined as a number, rather than an equivalence class, rests on (S8) pinning a version of the price: (S4) states Borel regularity for Voice paths only.*

Proof. Step 1 (the kernel is bounded). By (S8), $\pi \in [0, 1]$ and $p \in (0, 1)$, so their product satisfies

$0 \le h \le 1$ pointwise.

Step 2 (the kernel is a random variable). The posterior $\pi(\mathcal{I}_H) = \mathbb{P}(a = 1 \mid \mathcal{I}_H)$ is a version of a conditional probability and is $\mathcal{I}_H$-measurable, and so is the price $\mathbb{E}[Y \mid \mathcal{I}_H]$, whose version is pinned by (S8). That is what makes $h(\mathcal{I}_H)$ one well-defined random variable rather than an equivalence class from which the decomposition would have to choose. The entry probability is the composition of the posterior and the price with a continuous map, hence $\mathcal{I}_H$-measurable, and a product of measurable functions is measurable. With Step 1 the kernel is integrable.

Step 3 (the pointwise split). By Lemma 2 exactly one of $\mathbf{1}\{D = 1\}$ and $\mathbf{1}\{D = 0\}$ equals one at every sample point and the other equals zero, so $h = h\,\mathbf{1}\{D = 1\} + h\,\mathbf{1}\{D = 0\}$ pointwise.

Step 4 (integration). Each summand in Step 3 is bounded in absolute value by 1 by Step 1, hence integrable on the single probability space of (S1). Linearity of the integral gives $\mathbb{E}[h] = \mathbb{E}[h\,\mathbf{1}\{D = 1\}] + \mathbb{E}[h\,\mathbf{1}\{D = 0\}]$.

Step 5 (the interior case). Suppose $0 < \Omega < 1$. Both conditioning events have strictly positive probability, so the elementary conditional expectations $\mathbb{E}[h \mid D = 1] := \mathbb{E}[h\,\mathbf{1}\{D = 1\}]/\Omega$ and $\mathbb{E}[h \mid D = 0] := \mathbb{E}[h\,\mathbf{1}\{D = 0\}]/(1 - \Omega)$ are defined, and Step 1 bounds both by 1. This is the defining property of conditioning on an event of positive probability. Substituting into Step 4 gives $\mathbb{E}[h] = \Omega\,\mathbb{E}[h \mid D = 1] + (1 - \Omega)\,\mathbb{E}[h \mid D = 0]$.

Step 6 (scaling to premia). By (S9) the wedge Δ_m is finite and strictly positive. Multiplying Step 5 by Δ_m and reading off Δ_{act}, M_F and M_P gives (10).

Step 7 (the case $\Omega = 1$). Then $\mathbb{P}(D = 0) = 0$, and by Step 1 $|\mathbb{E}[h\,\mathbf{1}\{D = 0\}]| \le \mathbb{P}(D = 0) = 0$, so that term of Step 4 vanishes and $\mathbb{E}[h] = \mathbb{E}[h\,\mathbf{1}\{D = 1\}] = \Omega\,\mathbb{E}[h \mid D = 1] = \mathbb{E}[h \mid D = 1]$, the middle equality being Step 5's definition, available because $\Omega = 1 > 0$. Multiplying by Δ_m gives $\Delta_{\text{act}} = M_F$.

Step 8 (at $\Omega = 1$ the pooled average is not determined by the law). Step 5's definition does not apply to M_P, since it would divide by $\mathbb{P}(D = 0) = 0$. Conditioning on the generated σ-algebra supplies no value either. Let $\mathcal{D} := \sigma(D)$, whose atoms are $\{D = 1\}$ and $\{D = 0\}$, and let x be any real number. The function

$$\Delta_m\,\mathbb{E}[h \mid D = 1]\,\mathbf{1}\{D = 1\} \;+\; x\,\mathbf{1}\{D = 0\} \tag{11}$$

is $\mathcal{D}$-measurable and integrates to $\Delta_m\mathbb{E}[h\,\mathbf{1}_A]$ over every $A \in \mathcal{D}$, because two such functions with different values of x differ only on the null set $\{D = 0\}$. Every real x therefore yields a version of $\Delta_m\mathbb{E}[h \mid \mathcal{D}]$, so no value of M_P is determined by the law, and the statement at $\Omega = 1$ is Step 7's one-term equation.

Step 9 (the case $\Omega = 0$). Mirror Steps 7 and 8 with the cells exchanged: $|\mathbb{E}[h\,\mathbf{1}\{D = 1\}]| \le \mathbb{P}(D = 1) = 0$, so $\mathbb{E}[h] = \mathbb{E}[h \mid D = 0]$ and $\Delta_{\text{act}} = M_P$, while M_F is left undetermined by the argument of Step 8. □

The identity is tied to this partition. The identity (10) is tied to the disclosure partition and to no other. The flagged weight factors as $\Omega = \mathbb{P}(a = 1)\,\omega_a$ with $\omega_a = \mathbb{P}(D = 1 \mid a = 1)$, so $\Omega \le \mathbb{P}(a = 1)$, strictly when $\mathbb{P}(a = 1) > 0$ and $\omega_a < 1$. Averaging the kernel over $\{a = 1\}$ and $\{a = 0\}$ therefore gives a different decomposition, with the engaged but unflagged histories sitting in the pooled cell of (10).

The factorisation

Write $\mathcal{S} = |\partial_\kappa \Delta_{\text{act}}|$ for the liquidity sensitivity of the premium and $\mathcal{S}_P = |\partial_\kappa M_P|$ for the liquidity sensitivity of the pooled cell.
Label: PROVED.

Proposition 1 (Factorisation of the liquidity sensitivity)**.** *Assume (S1)–(S11), the partition of Lemma 2, the two-cell decomposition of Lemma 3, an interior partition $0 < \Omega < 1$ at the policy under consideration, the invariance of the flagged endpoint in κ at fixed policies, and, as an explicit hypothesis, differentiability of $\kappa \mapsto M_P$. Then*

$$\mathcal{S}(\kappa, \tau, T) \;=\; \big(1 - \Omega(\tau, T)\big)\, \mathcal{S}_P(\kappa, \tau, T), \tag{12}$$

exactly. The same factor scales the total variation of Δ_{act} over any grid in κ, and there no differentiability is used.

Proof. Step 1 (the two-cell identity). The partition is interior at the policy considered, so Lemma 3 applies in its non-degenerate branch and (10) holds at (κ, τ, T), with M_F and M_P the two cell averages.

Step 2 (the flagged cell does not move with κ). At the fixed cutoff and execution policies of (S11), the flagged posterior, the flagged price, the entry probability and M_F are invariant to κ, by the flagged-cell invariance assumed in the statement. Hence $\partial_\kappa M_F = 0$, and indeed M_F is one and the same number at any two values of κ.

Step 3 (the flagged weight does not move with κ either). This is derived, not assumed. By Step 4 of Lemma 2, $\{D = 1\}$ is the finite union (9), each of whose ingredients is a function of the signal alone once the cutoff policy is fixed. By (S7) the executed path $B_j(s, d)$ is a function of (j, s, d) alone, so no realised order flow and no realised price enters it, and with the policies frozen by (S11) the indicator D is a function of s alone. The law of s carries no κ: by (S1) the signal is $v + \varepsilon$, and by (S10) the laws of v and ε do not involve κ. Indeed κ enters the primitives in one place only, the law of the noise mark z_d, which reaches observed pooled order flow $X_d = q_{jd} + z_d$ and reaches nothing that D depends on. Hence $\Omega = \mathbb{P}(D = 1)$ is literally the same number at every κ, and in particular $\partial_\kappa \Omega = 0$. This step consumes the freezing of the cutoff vector in (S11), and no property of the disclosure rule beyond the partition.

Step 4 (differentiating in κ). Two of the three factors on the right of (10) are constant in κ: M_F by Step 2 and Ω by Step 3. Neither of those steps uses the present one, so nothing here is circular. With those factors constant, $\kappa \mapsto \Delta_{\text{act}}$ is an affine image of $\kappa \mapsto M_P$, which the statement's differentiability hypothesis makes differentiable; hence $\partial_\kappa \Delta_{\text{act}}$ exists and equals $(1 - \Omega)\, \partial_\kappa M_P$. Written with one term per factor that could carry κ, the same computation reads

$$\partial_\kappa \Delta_{\text{act}} \;=\; \Omega\, \partial_\kappa M_F \;+\; (1 - \Omega)\, \partial_\kappa M_P \;+\; (\partial_\kappa \Omega)\, (M_F - M_P), \tag{13}$$

the three terms being the flagged cell's own motion, the pooled cell's own motion, and the reallocation of mass between the cells. Step 2 kills the first and Step 3 the third. The third deletion assumes nothing about $M_F - M_P$, which is neither small nor signed here; the term goes because its coefficient is zero.

Step 5 (the factorisation). Since $0 < \Omega < 1$ we have $1 - \Omega \in (0,1)$ and $|1-\Omega| = 1 - \Omega$. Taking absolute values in the conclusion of Step 4 gives (12), exactly. Step 3 licenses writing the argument of Ω as (τ, T) rather than (κ, τ, T), and that convention is kept from here on. Two consequences are worth recording. First, $\mathcal{S} > 0$ exactly when $\mathcal{S}_P > 0$. Second, when $\mathcal{S}_P > 0$ the map $x \mapsto |x|$ is differentiable at $\partial_\kappa M_P \neq 0$, so $\mathcal{S}_P$ inherits whatever differentiability $\partial_\kappa M_P$ has. That transfer creates no differentiability; it moves it.

Step 6 (the aggregate form, with no differentiability used). Fix any grid $\kappa_0 < \kappa_1 < \cdots < \kappa_n$ inside the maintained parameter set, with the partition interior at each node. Applying Step 1 at κ_i and at κ_{i+1}, and using Steps 2 and 3 in their constancy form, that M_F and Ω are one and the same number at the two nodes, which a vanishing derivative at a single κ would not deliver, gives $\Delta_{\text{act}}(\kappa_{i+1}) - \Delta_{\text{act}}(\kappa_i) = (1-\Omega)\big[M_P(\kappa_{i+1}) - M_P(\kappa_i)\big]$. Summing absolute values,

$$\sum_{i=0}^{n-1} \big|\Delta_{\text{act}}(\kappa_{i+1}) - \Delta_{\text{act}}(\kappa_i)\big| \;=\; (1-\Omega) \sum_{i=0}^{n-1} \big|M_P(\kappa_{i+1}) - M_P(\kappa_i)\big|, \tag{14}$$

which is the total-variation clause, finite differences throughout. The same argument runs for any aggregator of the increment vector that is positively homogeneous of degree one, such as total variation, mean absolute slope, or the supremum of the absolute increments, because $1 - \Omega$ is one nonnegative scalar common to every increment. A measurement convention that reports κ-sensitivity as a total variation over a grid rather than as a pointwise derivative is then measuring the same functional. □

Label: PROVED.

Lemma 4 (The weight leg at the threshold margin)**.** *Fix the liquidity κ, the window margin T, and the plan and cutoff policies. Compare two threshold margins with $b_0 < \tau' < \tau$, so that τ' is the tighter threshold, and assume $\Omega(\tau', T) < 1$. The configuration $b_0 < \tau'$ is what puts the clock equivalence in force at both thresholds, together with Voice date-monotonicity (standing condition (S4)) and the restriction that only Voice plans cross (standing condition (S5)). Then*

- *(i) the flagged cells are nested, $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$, and every history the tighter threshold newly flags carries the engagement flag $a = 1$;*
- *(ii) the flagged weight rises, $\Omega(\tau', T) \geq \Omega(\tau, T)$, so the weight leg*

$$W_\tau \;=\; \frac{1 - \Omega(\tau', T)}{1 - \Omega(\tau, T)} \tag{15}$$

  is defined and attenuates: $0 < W_\tau \leq 1$;
- *(iii) if in addition $\Omega(\tau, T) > 0$, $\mathcal{S}_P(\kappa, \tau, T) > 0$, and the hypotheses of Proposition 1 hold at both thresholds, the sensitivity ratio at this margin factors into the weight leg and the composition leg $C_\tau = \mathcal{S}_P(\kappa, \tau', T)/\mathcal{S}_P(\kappa, \tau, T)$,*

$$\frac{\mathcal{S}(\kappa, \tau', T)}{\mathcal{S}(\kappa, \tau, T)} \;=\; W_\tau\, C_\tau. \tag{16}$$

Proof. Fixed policies hold the map from the signal to the plan, the executed stake path of each plan, and the law of the signal common to the two rules. The disclosure rule writes the stake path with no threshold argument, while the crossing date, the filing date, the stake at filing and the disclosure indicator each carry one. The only objects that move when the threshold moves are therefore those four, and every comparison below is made at one and the same plan, one and the same path, and one and the same signal law.

The product form of the disclosure indicator. For every plan j, every signal s, and each of the two thresholds,

$$D_j(s;\tau,T) \;=\; \mathbf{1}\{a_j = 1\} \cdot \mathbf{1}\{B_j(s, H-T) \geq \tau\}. \tag{17}$$

If $a_j = 0$ both sides vanish, the left side because engagement is a conjunct of disclosure and the right side through its first factor. If $a_j = 1$ the left side is the event that the filing lands by the horizon: the filing date is the crossing date plus the window margin, and with a finite window margin the filing landing by the horizon already forces the crossing to be finite, so the requirement that a crossing occur adds nothing. The clock equivalence then converts the filing landing by the horizon into the stake T rounds before the horizon having reached the threshold, which is the second factor. Because $b_0 < \tau' < \tau$, the crossing is the first passage at each of the two thresholds and the equivalence is available at both.

The inclusion. Take any control-node history that the looser threshold flags, $D_j(s;\tau,T) = 1$. By (17) this says $a_j = 1$ and $B_j(s, H-T) \geq \tau$. Since $\tau' < \tau$, the same real number satisfies $B_j(s, H-T) \geq \tau > \tau'$, and reading (17) at τ' for the same plan and the same signal gives $D_j(s;\tau',T) = 1$. The number $B_j(s, H-T)$ that appears in the two comparisons is one number, and this is where fixed policies are consumed: were the path free to move with the threshold, the comparison at τ' would be about a different path. The history was arbitrary, so $\mathcal{C}_F(\tau,T) \subseteq \mathcal{C}_F(\tau',T)$, and taking complements in the exclusive and exhaustive partition gives $\mathcal{C}_P(\tau',T) \subseteq \mathcal{C}_P(\tau,T)$: a tighter threshold can only shrink the pooled cell. Every newly flagged history satisfies $D(\tau') = 1$, and by (17) that carries $a = 1$ as a conjunct, so the engagement flag is identically one on the newly flagged set. Nestedness is a conclusion here rather than a hypothesis. This is clause (i).

The weight leg. The joint law of the primitives is itself a primitive and does not depend on the threshold, which enters only through the flagged event. Monotonicity of a probability measure applied to the inclusion gives

$$\Omega(\tau',T) \;=\; \mathbb{P}\big(\mathcal{C}_F(\tau',T)\big) \;\geq\; \mathbb{P}\big(\mathcal{C}_F(\tau,T)\big) \;=\; \Omega(\tau,T),$$

so $1 - \Omega(\tau',T) \leq 1 - \Omega(\tau,T)$. The hypothesis $\Omega(\tau',T) < 1$ makes the numerator of (15) strictly positive, and the same hypothesis with the inequality just established gives $\Omega(\tau,T) \leq \Omega(\tau',T) < 1$, so the denominator is strictly positive as well. The quotient is therefore defined, and dividing a weakly smaller strictly positive number by a strictly positive one gives $0 < W_\tau \leq 1$. This is clause (ii).

The ratio identity. With $\Omega(\tau,T) > 0$ the weight-leg inequality puts both flagged weights strictly inside the unit interval, so Proposition 1 applies at each threshold and the noise sensitivity of the expected engagement premium is the pooled share times the pooled sensitivity,

$$\mathcal{S}(\kappa,\tau,T) \;=\; \big(1 - \Omega(\tau,T)\big)\,\mathcal{S}_P(\kappa,\tau,T), \qquad \mathcal{S}(\kappa,\tau',T) \;=\; \big(1 - \Omega(\tau',T)\big)\,\mathcal{S}_P(\kappa,\tau',T).$$

The flagged weight does not move with κ, which is what licenses writing it as $\Omega(\cdot, T)$ with no liquidity argument on either side. Since $1 - \Omega(\tau, T) > 0$ and $\mathcal{S}_P(\kappa, \tau, T) > 0$, the denominator $\mathcal{S}(\kappa, \tau, T)$ is strictly positive and the quotient of the two displays is defined. Dividing and grouping the two factors gives (16), which is clause (iii). $\square$

The weight leg at this margin is signed outright, and it is signed by the geometry of the flagged set alone: no restriction on the pooled experiment and no restriction on the noise-trading intensity enters clauses (i) and (ii). What clause (iii) then supplies is the exact form in which the composition leg has to be signed: $W_\tau C_\tau \leq 1$ if and only if $C_\tau \leq 1/W_\tau$, and by clause (ii) that ceiling is at least one.
Label: PROVED.

Theorem 1 (The clock dial)**.** *Fix the liquidity κ, the threshold margin τ, and the plan and cutoff policies. Compare two window margins $T' < T$, so that T' is the shorter clock. Assume $b_0 < \tau$, that Voice stake paths are weakly increasing in the calendar date (standing condition (S4)), and that only Voice plans cross (standing condition (S5)), which is the configuration that puts the clock equivalence in force at both windows. Assume $0 < \Omega(\tau, T) < 1$ and $0 < \Omega(\tau, T') < 1$, and $\mathcal{S}_P(\kappa, \tau, T) > 0$. Assume that the hypotheses of Proposition 1 hold at both clocks. Write*

$$W_T \;=\; \frac{1 - \Omega(\tau, T')}{1 - \Omega(\tau, T)}, \qquad C_T \;=\; \frac{\mathcal{S}_P(\kappa, \tau, T')}{\mathcal{S}_P(\kappa, \tau, T)}. \tag{18}$$

Then

- *(i) the weight leg attenuates: $0 < W_T \leq 1$;*
- *(ii) the sensitivity ratio factors into the two legs,*

$$\frac{\mathcal{S}(\kappa, \tau, T')}{\mathcal{S}(\kappa, \tau, T)} \;=\; W_T\, C_T; \tag{19}$$

- *(iii) the shorter clock lowers noise sensitivity exactly when the product of the two legs is at most one,*

$$\mathcal{S}(\kappa, \tau, T') \;\leq\; \mathcal{S}(\kappa, \tau, T) \qquad \Longleftrightarrow \qquad W_T\, C_T \;\leq\; 1. \tag{20}$$

Proof. Fixed policies hold the map from the signal to the plan, the executed stake path of each plan, and the law of the signal common to the two rules, so every object below is compared at one and the same plan and one and the same signal law.

The weight leg. Let j be a Voice plan and s a signal whose history is flagged under the longer clock, $D_j(s; \tau, T) = 1$. By the clock equivalence, the filing lands by the horizon exactly when the stake T rounds before the horizon has reached the threshold, so $B_j(s, H - T) \geq \tau$. Since $T' < T$ gives $H - T' > H - T$, and since the stake path of a Voice plan is weakly increasing in the round, $B_j(s, H - T') \geq B_j(s, H - T) \geq \tau$. The engagement flag $a_j = 1$ does not move with the clock. Reading the clock equivalence in the other direction at the shorter clock gives $D_j(s; \tau, T') = 1$. Only Voice plans reach the threshold, so no other history has to be checked, and the flagged cell at the longer clock is contained in the flagged cell at the shorter clock. Taking probabilities under the common signal law, $\Omega(\tau, T') \geq \Omega(\tau, T)$, so $1 - \Omega(\tau, T') \leq 1 - \Omega(\tau, T)$. Both sides are strictly positive because $\Omega < 1$ at each clock, so the quotient in (18) is defined and $0 < W_T \leq 1$. This is clause (i).

The ratio identity. By Proposition 1 at both clocks, which the statement assumes, the noise sensitivity of the expected engagement premium is the pooled share times the pooled sensitivity. The factorisation hypotheses at the shorter clock, differentiability of $\kappa \mapsto M_P(\kappa, \tau, T')$ included, make $\mathcal{S}_P(\kappa, \tau, T')$ exist, so C_T in (18) is defined. At each clock,

$$\mathcal{S}(\kappa, \tau, T) \ = \ \big(1 - \Omega(\tau, T)\big)\, \mathcal{S}_P(\kappa, \tau, T), \qquad \mathcal{S}(\kappa, \tau, T') \ = \ \big(1 - \Omega(\tau, T')\big)\, \mathcal{S}_P(\kappa, \tau, T').$$

The flagged weight does not move with κ, which is what licenses writing it as $\Omega(\tau, \cdot)$ with no liquidity argument on either side. Since $1 - \Omega(\tau, T) > 0$ and $\mathcal{S}_P(\kappa, \tau, T) > 0$, the denominator $\mathcal{S}(\kappa, \tau, T)$ is strictly positive and the quotient of the two displays is defined. Dividing and grouping the two factors gives (19), which is clause (ii).

The equivalence. Multiplying an inequality by the strictly positive number $\mathcal{S}(\kappa, \tau, T)$ preserves it in both directions, so $\mathcal{S}(\kappa, \tau, T') \leq \mathcal{S}(\kappa, \tau, T)$ holds exactly when $\mathcal{S}(\kappa, \tau, T')/\mathcal{S}(\kappa, \tau, T) \leq 1$, and by (19) that quotient is $W_T C_T$. This is (20) and clause (iii). $\square$

Because $W_T > 0$, the criterion (20) can be read as a ceiling on the composition leg: the shorter clock lowers noise sensitivity exactly when

$$C_T \ \leq \ \frac{1}{W_T} \ = \ \frac{1 - \Omega(\tau, T)}{1 - \Omega(\tau, T')}.$$

By clause (i) that ceiling is at least one, so $C_T \leq 1$ is sufficient for attenuation at the clock margin.

2 Order size two: the pooled experiment and the threshold dial

Setting and notation

Trading runs over rounds $d = 0, \ldots, H$. Noise marks $(z_d)_{d=0}^{H}$ are independent and identically distributed across rounds and independent of the blockholder's type θ (and thus of the signal s and mark path $(m_d)_{d=0}^{H}$). In each round, noise takes the values $-\bar{z}$, 0 and $+\bar{z}$ with probabilities $\kappa/2$, $1 - \kappa$ and $\kappa/2$. The blockholder's order while she is building the stake is two noise lumps, so the pooled order mark is $q_{jd}(s) \in \{0, 2\bar{z}\}$ and pooled order flow is $X_d = q_{jd}(s) + z_d$. The support of X_d is $\{-\bar{z}, 0, \bar{z}, 2\bar{z}, 3\bar{z}\}$: a round in which the blockholder buys, an *active* round, produces $X_d \in \{\bar{z}, 2\bar{z}, 3\bar{z}\}$, and a round in which she does not, an *idle* round, produces $X_d \in \{-\bar{z}, 0, \bar{z}\}$. The two supports meet at $\bar{z}$ and nowhere else. Write

$$\varepsilon \ := \ \kappa/2 \ \in \ (0, \tfrac{1}{2}) \qquad \text{for } \kappa \in (0, 1). \tag{21}$$

Policies are fixed throughout: the plan menu, the cutoff vector and the execution paths are held at one setting while the disclosure rule (τ, T) moves. Let θ denote the blockholder's type, the pair of plan and signal, and let

$$m_d(\theta) \ \in \ \{0,\, 2\bar{z}\}, \qquad d = 0, \ldots, H,$$

be its mark path, equal to $2\bar{z}$ on the rounds in which the plan buys and 0 elsewhere. The mark path is a deterministic function of the type and takes finitely many values. The disclosure indicator $D(\theta) = \mathbf{1}\{a(\theta) = 1,\ c(\theta;\tau) + T \le H\}$ is a function of the type alone, because the stake path does not respond to realised order flow and the crossing date reads off that path. Write ρ for the prior law of θ, $\Omega(\tau, T) = \mathbb{P}(D = 1)$ for the flagged weight, and, when $\Omega(\tau, T) < 1$,

$$\rho_P^{\tau,T} \;:=\; \rho(\,\cdot \mid D = 0\,)$$

for the pooled type law, which does not depend on κ.

The kernel is $h(\mathcal{I}) = \pi(\mathcal{I})\, p(\mathcal{I})$, with π the engagement posterior and p the entry probability. The price and the entry probability at an information set are pinned by the posterior engagement share π and the posterior mean $\hat{v}$ of the fundamental, so

$$h(\nu) \;=\; \mathsf{h}\big(\pi(\nu),\, \hat{v}(\nu)\big), \tag{22}$$

where ν is the posterior over types and $\pi(\nu)$, $\hat{v}(\nu)$ are both linear in ν. The map h itself carries no κ: κ enters $h(\nu)$ only through the posterior ν. Lemma 7(a) rests on that, treating each $\mathcal{G}(S)$ as a κ-free number. A curvature statement about h is read as a statement about h on the convex hull of the pairs $(\pi, \hat{v})$ the pooled cell generates. The premium objects are $\Delta_{\text{act}} = \Delta_m \mathbb{E}[h(\mathcal{I}_H)]$, $M_F = \Delta_m \mathbb{E}[h \mid D = 1]$, $M_P = \Delta_m \mathbb{E}[h \mid D = 0]$, and the sensitivities are $\mathcal{S} = |\partial_\kappa \Delta_{\text{act}}|$ and $\mathcal{S}_P = |\partial_\kappa M_P|$. For a threshold pair $\tau' < \tau$ at a common window,

$$W_\tau \;=\; \frac{1 - \Omega(\tau', T)}{1 - \Omega(\tau, T)}, \qquad C_\tau \;=\; \frac{\mathcal{S}_P(\tau', T)}{\mathcal{S}_P(\tau, T)}.$$

For a set of rounds S, the *S-record* is the pair $\big(S,\ (m_e(\theta))_{e \in S}\big)$; Lemma 5(b) shows that the pooled history determines it. Finally, call an event $\mathcal{A}$ a *cell event* when it is determined by the type alone, so that conditioning on it leaves the noise law untouched. The pooled cell $\{D = 0\}$ is a cell event, and so is the event that the filing has not landed by a date d. For a cell event $\mathcal{A}$ with $\mathbb{P}(\mathcal{A}) > 0$, write $\rho_{\mathcal{A}} := \rho(\,\cdot \mid \mathcal{A}\,)$.

Statements

Label: PROVED.

Lemma 5 (Erasure form of the pooled experiment)**.** *Fix a policy (τ, T), a depth $d \le H$, and $\kappa \in (0, 1)$. Assume the noise marks $(z_e)_{e \le d}$ are i.i.d. and independent of θ. Let*

$$R_d \;:=\; \{\, e \le d \;:\; X_e \ne \bar{z} \,\}$$

be the set of rounds whose flow differs from one lump. Then:

- *(a) $\mathbb{P}(X_e = \bar{z} \mid \theta) = \varepsilon$ for every type and every round, so the value $\bar{z}$ carries likelihood ratio one across types and its probability rises with κ; consequently R_d is independent of θ and each round belongs to it independently with probability $1 - \varepsilon$;*
- *(b) on $\{R_d = S\}$ the flow determines $m_e(\theta)$ for every $e \in S$, and conditional on the S-record the law of $X_{0:d}$ does not depend on θ;*

(c) *consequently the S-record is sufficient: for every cell event $\mathcal{A}$ with $\mathbb{P}(\mathcal{A}) > 0$, the posterior after observing $X_{0:d}$ on $\mathcal{A}$ is*

$$\nu_{0:d} \;=\; \rho_{\mathcal{A}}\big(\cdot \mid (m_e)_{e\in R_d}\big),$$

and in particular, when $\Omega(\tau, T) < 1$, the pooled posterior at the control node takes this form with $\mathcal{A} = \{D = 0\}$ and $d = H$.

Label: PROVED.

Lemma 6 (Liquidity garbles the pooled experiment)**.** *Fix a policy and a depth $d \le H$, let $0 < \kappa < \kappa' < 1$, and assume the noise marks $(z_e)_{e\le d}$ are i.i.d. and independent of θ. There is a Markov kernel Λ from histories to histories, not depending on the type, such that for every type θ the image under Λ of the law of $X_{0:d}$ at κ is its law at κ'. One such kernel deletes each element of R_d independently with probability $(\kappa' - \kappa)/(2 - \kappa)$, emits $\bar{z}$ on every deleted round and on every round outside R_d, and redraws the flow on the surviving rounds from the κ' conditional law given the mark. The pooled experiment at κ' is therefore a garbling of the pooled experiment at κ, and the law of the pooled posterior at κ' is a mean preserving contraction of its law at κ, and when $\Omega(\tau, T) < 1$, both have mean $\rho_P^{\tau,T}$.*

Label: PROVED.

Lemma 7 (Exact liquidity representation of the pooled premium)**.** *Fix a policy (τ, T), a cell event $\mathcal{A}$ of positive prior mass, a depth $d \le H$, and assume the noise marks $(z_e)_{e\le d}$ are i.i.d. and independent of θ. For $S \subseteq \{0, \ldots, d\}$ define*

$$\mathcal{G}^{\mathcal{A},d}(S) \;:=\; \mathbb{E}_{\rho_{\mathcal{A}}}\Big[\, h\big(\rho_{\mathcal{A}}(\cdot \mid (m_e)_{e\in S})\big) \Big], \tag{23}$$

a number that does not depend on κ. The three parts below are written for the pooled cell at the control node, $\mathcal{A} = \{D = 0\}$ and $d = H$, where $\mathcal{G}_{\tau,T} := \mathcal{G}^{\{D=0\},H}$ and $\Omega(\tau, T) < 1$; each holds verbatim for any cell event and any depth $d \le H$, with H replaced by d throughout.

(a) Representation. *For every $\kappa \in (0, 1)$,*

$$M_P(\tau, T; \kappa) \;=\; \Delta_m \sum_{S\subseteq\{0,\ldots,H\}} (1-\varepsilon)^{|S|}\, \varepsilon^{\,H+1-|S|}\, \mathcal{G}_{\tau,T}(S). \tag{24}$$

(b) Derivative. *With the κ-free coefficients*

$$c_k(\tau, T) \;:=\; \sum_{d=0}^{H} \sum_{\substack{S\subseteq\{0,\ldots,H\}\setminus\{d\}\\ |S|=k}} \Big[\, \mathcal{G}_{\tau,T}(S \cup \{d\}) - \mathcal{G}_{\tau,T}(S) \Big], \qquad k = 0, \ldots, H, \tag{25}$$

the pooled premium is differentiable in κ on $(0, 1)$ and

$$\partial_\kappa M_P(\tau, T; \kappa) \;=\; -\frac{\Delta_m}{2}\, \mathcal{W}_{\tau,T}(\kappa), \qquad \mathcal{W}_{\tau,T}(\kappa) \;:=\; \sum_{k=0}^{H} (1-\varepsilon)^k \varepsilon^{\,H-k}\, c_k(\tau, T). \tag{26}$$

(c) Sign. *If* h *of* (22) *is concave on the convex hull of the pairs* $(\pi, \hat{v})$ *the pooled cell generates, then every increment in* (25) *is at most zero, hence* $c_k(\tau, T) \le 0$ *for every* k *and* $\partial_\kappa M_P \ge 0$ *at every* $\kappa \in (0,1)$*. If* h *is convex there, all three inequalities reverse. The pooled expectation of the kernel is therefore monotone in* κ*, in the direction the sign of the chord of* h *gives, and the same holds at every depth* $d \le H$*.*

Condition 1 (Revelation dominance at a threshold pair)**.** *Let* $b_0 < \tau' < \tau$ *at a common window* T*, and let* $\mathcal{W}$ *be as in* (26)*. The pair satisfies revelation dominance at* κ *if*

$$\big|\mathcal{W}_{\tau',T}(\kappa)\big| \;\le\; \big|\mathcal{W}_{\tau,T}(\kappa)\big|. \tag{27}$$

Labels: PROVED and NUMERICAL. Parts (A) and (B) and the implication in part (C) are PROVED; Condition 1, and with it the conclusion of part (C), is NUMERICAL on the grid Remark 3 names.

Theorem 2 (The threshold dial at fixed policies)**.** *Fix the plan and cutoff policies and a window* T*, and take* $b_0 < \tau' < \tau$ *with* $\Omega(\tau, T) > 0$*,* $\Omega(\tau', T) < 1$ *and* $\mathcal{S}_P(\tau, T) > 0$*; part (B) then puts* $\Omega(\tau, T) \le \Omega(\tau', T) < 1$*, so both pooled cells carry positive mass. Assume the two cell decomposition* $\Delta_{\text{act}} = \Omega M_F + (1-\Omega) M_P$ *and the* κ*-invariance of the flagged average* M_F*, each with its own hypotheses. Then:*

(A) Factorisation. $\mathcal{S} = (1-\Omega)\,\mathcal{S}_P$*, exactly, at every* $\kappa \in (0,1)$*.*

(B) Weight leg. $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$ *and every newly flagged history is generated by a Voice plan, so* $\Omega(\tau', T) \ge \Omega(\tau, T)$ *and* $W_\tau \in [0,1]$*, with equality when no history is reclassified (that is, up to a null set of histories). This leg carries no condition beyond the standing ones.*

(C) Composition leg and the dial. *Under Condition 1 at the pair and at* κ*,* $C_\tau \le 1$*, and therefore*

$$\frac{\mathcal{S}(\tau', T)}{\mathcal{S}(\tau, T)} \;=\; W_\tau\, C_\tau \;\le\; 1. \tag{28}$$

Equality holds when no history is reclassified (up to a null set of histories), since the two rules then induce the same partition almost surely and both factors are one.

Remark 1 (A κ-free sufficient form of Condition 1)**.** *The coefficients* $c_k(\tau, T)$ *of* (25) *do not depend on* κ*, so Condition 1 is a comparison of two vectors in* $\mathbb{R}^{H+1}$*. If the* $H+1$ *numbers* $c_0(\tau, T), \ldots, c_H(\tau, T)$ *share one sign and*

$$|c_k(\tau', T)| \;\le\; |c_k(\tau, T)| \qquad \textit{for every } k = 0, \ldots, H, \tag{29}$$

then Condition 1 holds at every $\kappa \in (0,1)$ *at once.*

Remark 2 (What the coefficients are)**.** *By Lemma 5 the* S*-record partitions the pooled type set into the level sets of the restricted mark path* $\theta \mapsto (m_e(\theta))_{e \in S}$*, so* $\mathcal{G}_{\tau,T}(S)$ *is a finite sum over those level sets of the cell mass times the kernel at the cell posterior. Each* $c_k(\tau, T)$ *is therefore a closed form expression in the pooled type weights, computable without reference to* κ *and without enumerating order flow.*

Proofs

Proof of Lemma 5. Condition on a type θ and a round $e \le d$. If the round is active the mark is $2\bar{z}$ and X_e takes the values $\bar{z}$, $2\bar{z}$, $3\bar{z}$ with probabilities $\kappa/2$, $1-\kappa$, $\kappa/2$. If the round is idle the mark is zero and X_e takes the values $-\bar{z}$, 0, $\bar{z}$ with the same three probabilities in the same order. In both cases

$$\mathbb{P}(X_e = \bar{z} \mid \theta) \;=\; \kappa/2 \;=\; \varepsilon,$$

which proves the first claim of part (a); the noise draws are independent across rounds and independent of the type, so the indicators $\mathbf{1}\{e \in R_d\}$, $e \le d$, are independent Bernoulli variables with success probability $1-\varepsilon$, independent of θ. That is part (a).

For part (b), the two supports meet only at $\bar{z}$: an active round produces $X_e \in \{2\bar{z}, 3\bar{z}\}$ and an idle round produces $X_e \in \{-\bar{z}, 0\}$ whenever $X_e \ne \bar{z}$. So on $\{e \in R_d\}$ the flow determines the mark. It remains to check that nothing else is carried. Conditional on the round being active and revealing,

$$\mathbb{P}(X_e = 2\bar{z} \mid \text{active}, e \in R_d) = \frac{1-\kappa}{1-\varepsilon}, \qquad \mathbb{P}(X_e = 3\bar{z} \mid \text{active}, e \in R_d) = \frac{\varepsilon}{1-\varepsilon},$$

and conditional on the round being idle and revealing the values 0 and $-\bar{z}$ carry those same two probabilities. The two conditional laws are the same pair of numbers, so once the mark is given, which of its two revealing values was realised is a draw whose law does not depend on the type. Rounds outside R_d carry the single value $\bar{z}$ and no residual randomness at all. Combining across rounds by independence, the conditional law of $X_{0:d}$ given the S-record does not depend on θ, which is part (b).

Part (c) is the sufficiency conclusion. Write $\mathcal{R}$ for the S-record. Since the conditional law of $X_{0:d}$ given $(\mathcal{R}, \theta)$ does not depend on θ, θ and $X_{0:d}$ are conditionally independent given $\mathcal{R}$, and Bayes' rule gives $\mathbb{P}(\theta \in \cdot \mid X_{0:d}, D = 0) = \mathbb{P}(\theta \in \cdot \mid \mathcal{R}, D = 0)$. The event $\{D = 0\}$ is a type event, so conditioning on it is conditioning ρ down to $\rho_P^{\tau,T}$, and the displayed posterior is the one stated. $\square$

Proof of Lemma 6. Let $\varepsilon = \kappa/2 < \varepsilon' = \kappa'/2$ and put

$$\delta \;:=\; \frac{\varepsilon' - \varepsilon}{1-\varepsilon} \;=\; \frac{\kappa' - \kappa}{2-\kappa} \;\in\; (0,1).$$

Define Λ on a κ-history as follows. First read off the record $(R_d, (m_e)_{e \in R_d})$, which Lemma 5(b) allows without knowing the type. Second, delete each element of R_d independently with probability δ, obtaining R'_d. Third, emit the flow value $\bar{z}$ on every round outside R'_d, and on every round $e \in R'_d$ emit one of the two revealing values consistent with m_e, drawn with probabilities $(1-\kappa')/(1-\varepsilon')$ for the zero noise value and $\varepsilon'/(1-\varepsilon')$ for the other. The kernel reads only the history, never the type.

Fix a type. Under the κ law each round lies in R_d independently with probability $1-\varepsilon$, so under Λ each round lies in R'_d independently with probability

$$(1-\varepsilon)(1-\delta) \;=\; (1-\varepsilon) - (\varepsilon' - \varepsilon) \;=\; 1 - \varepsilon'.$$

On the rounds of R'_d the emitted marks are the type's own marks, and the revealing value is drawn from the κ' conditional law computed in the proof of Lemma 5. By that lemma's factorisation across rounds, the emitted history has exactly the κ' law given the type. So Λ carries the κ experiment to the κ' experiment and the second is a garbling of the first.

For the final claim, run Λ on the κ history to build a joint law of the type, the κ history and the κ' history. Because Λ reads only the history, those three form a Markov chain in that order, so

$$\mathbb{P}\big(\theta \in \cdot \;\big|\; X^{\kappa'}_{0:d}\big) \;=\; \mathbb{E}\Big[\mathbb{P}\big(\theta \in \cdot \;\big|\; X^{\kappa}_{0:d}\big) \;\Big|\; X^{\kappa'}_{0:d}\Big].$$

The κ' posterior is therefore a conditional expectation of the κ posterior, so the law of the first is a mean preserving contraction of the law of the second, and when $\Omega(\tau, T) < 1$, both have mean $\rho_P^{\tau,T}$. □

Proof of Lemma 7. Part (a). By Lemma 5(a) the revealed set R_H has the product Bernoulli law $\mathbb{P}(R_H = S) = (1-\varepsilon)^{|S|}\varepsilon^{H+1-|S|}$, and that law is the same under ρ and under $\rho_P^{\tau,T}$, because $\{D = 0\}$ is a type event and R_H is independent of the type. By Lemma 5(c) the pooled posterior on $\{R_H = S\}$ is $\rho_P^{\tau,T}(\cdot \mid (m_e)_{e\in S})$. Conditioning on R_H and applying the tower property,

$$\mathbb{E}\big[h(\nu_{0:H}) \mid D = 0\big] \;=\; \sum_S \mathbb{P}(R_H = S)\, \mathbb{E}_{\rho_P^{\tau,T}}\Big[h\big(\rho_P^{\tau,T}(\cdot \mid (m_e)_{e\in S})\big)\Big] \;=\; \sum_S \mathbb{P}(R_H = S)\, \mathcal{G}_{\tau,T}(S),$$

and multiplying by Δ_m gives (24). Nothing in the argument used the depth or the particular cell event, so the same identity holds with H replaced by any $d \le H$ and $\{D = 0\}$ replaced by any cell event. The numbers $\mathcal{G}_{\tau,T}(S)$ are built from $\rho_P^{\tau,T}$ and the mark paths only, and neither depends on κ.

Part (b). Write $q := 1 - \varepsilon$ and $F(q) := \sum_S q^{|S|}(1-q)^{H+1-|S|}\mathcal{G}(S)$, the expectation of $\mathcal{G}(R_H)$ under the product Bernoulli measure with success probability q on each of the $H+1$ rounds. Read F as a function of $H+1$ separate success probabilities, one per round. It is affine in each of them, because the rounds enter the product measure independently, and its partial derivative in the d-th is $\mathbb{E}[\mathcal{G}(R_{-d} \cup \{d\})] - \mathbb{E}[\mathcal{G}(R_{-d})]$, with R_{-d} the random set on the other rounds. Setting every coordinate to q and summing the $H+1$ partial derivatives,

$$F'(q) \;=\; \sum_{d=0}^{H} \mathbb{E}_q\Big[\,\mathcal{G}\big(R_{-d} \cup \{d\}\big) - \mathcal{G}\big(R_{-d}\big)\Big],$$

Expanding the expectation over R_{-d} and collecting terms by $|S| = k$,

$$F'(q) \;=\; \sum_{d=0}^{H} \;\sum_{S \subseteq \{0,\dots,H\}\setminus\{d\}} q^{|S|}(1-q)^{H-|S|}\big[\mathcal{G}(S \cup \{d\}) - \mathcal{G}(S)\big] \;=\; \sum_{k=0}^{H} q^k(1-q)^{H-k} c_k(\tau, T).$$

Since $M_P = \Delta_m F(q)$ and $q = 1 - \kappa/2$ gives $dq/d\kappa = -1/2$, the chain rule yields (26) with $1 - \varepsilon = q$ and $\varepsilon = 1 - q$.

Part (c). Fix S and $d \notin S$, and write ν_S and $\nu_{S\cup\{d\}}$ for the two posteriors. The record on S is coarser than the record on $S \cup \{d\}$, so $\nu_S = \mathbb{E}[\nu_{S\cup\{d\}} \mid (m_e)_{e\in S}]$. Both π and $\hat{v}$ are linear in the posterior, so the pair $(\pi, \hat{v})$ evaluated at these two posteriors is an $\mathbb{R}^2$ valued martingale over the two records, and its values lie in the convex hull named in the statement. If h is concave there, the conditional Jensen inequality in $\mathbb{R}^2$ gives

$$\mathbb{E}\big[\mathsf{h}(\pi_{S\cup\{d\}}, \hat{v}_{S\cup\{d\}}) \bigm| (m_e)_{e\in S}\big] \;\le\; \mathsf{h}(\pi_S, \hat{v}_S) \quad \text{almost surely,}$$

and taking expectations, $\mathcal{G}(S \cup \{d\}) \le \mathcal{G}(S)$. Every increment in (25) is then at most zero, so $c_k \le 0$ for every k. The weights $(1-\varepsilon)^k \varepsilon^{H-k}$ are strictly positive on $\kappa \in (0,1)$ and $\Delta_m > 0$, so (26) gives $\partial_\kappa M_P \ge 0$ there. Reversing the curvature reverses each of the three steps. The depth played no role, so the statement holds at every $d \le H$. $\square$

Proof of Theorem 2. Part (A). At fixed policies the disclosure indicator is a function of the type alone: the plan fixes the stake path, the path fixes the crossing date, and the filing lands by the control date exactly when the crossing date is at most $H - T$. Liquidity enters only the noise law, which touches order flow and not the stake path, so Ω does not move with κ. The flagged average M_F does not move with κ by the flagged invariance assumed in the statement. Differentiating $\Delta_{\text{act}} = \Omega M_F + (1-\Omega) M_P$,

$$\partial_\kappa \Delta_{\text{act}} = (\partial_\kappa \Omega)(M_F - M_P) + \Omega\, \partial_\kappa M_F + (1-\Omega)\, \partial_\kappa M_P = (1-\Omega)\, \partial_\kappa M_P,$$

the first two terms vanishing term by term rather than by cancellation. Differentiability of M_P in κ is not a hypothesis here: Lemma 7(a) exhibits M_P as a polynomial in κ. Taking absolute values and using $1 - \Omega \ge 0$ gives $\mathcal{S} = (1-\Omega)\mathcal{S}_P$.

Part (B). Let a history be flagged at τ, so $a = 1$ and $c(\tau) + T \le H$, where $c(\tau) = \inf\{d : B(s,d) \ge \tau\}$. Since $\tau' < \tau$, every date at which the stake reaches τ is a date at which it reaches τ', so $c(\tau') \le c(\tau)$ and therefore $c(\tau') + T \le H$. The engagement flag is unchanged, so the history is flagged at τ' as well, and $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$. Disclosure implies engagement, so every history in the difference carries $a = 1$ and is generated by a Voice plan. Taking masses, $\Omega(\tau', T) \ge \Omega(\tau, T)$, and with $\Omega(\tau', T) < 1$,

$$W_\tau = \frac{1 - \Omega(\tau', T)}{1 - \Omega(\tau, T)} \in [0, 1],$$

with $W_\tau = 1$ exactly when the two flagged sets have the same mass, that is when no history is reclassified (almost surely). The argument used no property of the kernel.

Part (C). Both thresholds are evaluated at the same κ and at the same policies, so Lemma 7(b) applies at each and gives

$$\mathcal{S}_P(\tau, T) = \frac{\Delta_m}{2} \big|\mathcal{W}_{\tau,T}(\kappa)\big|, \qquad \mathcal{S}_P(\tau', T) = \frac{\Delta_m}{2} \big|\mathcal{W}_{\tau',T}(\kappa)\big|.$$

Since $\mathcal{S}_P(\tau, T) > 0$ the ratio C_τ is defined, and Condition 1 states exactly that its numerator is at most its denominator, so $C_\tau \le 1$. Part (A) applied at each threshold gives $\mathcal{S}(\tau', T)/\mathcal{S}(\tau, T) = W_\tau C_\tau$, and part (B) puts W_τ in $[0,1]$. The product of a number in $[0,1]$ with a number in $[0,1]$ is at most one, which is (28), with equality exactly when both factors are one. $\square$

Proof of Remark 1. Write $w_k = (1-\varepsilon)^k \varepsilon^{H-k} > 0$. By the triangle inequality and (29),

$$\big|\mathcal{W}_{\tau',T}(\kappa)\big| \;\le\; \sum_k w_k\, |c_k(\tau', T)| \;\le\; \sum_k w_k\, |c_k(\tau, T)|.$$

If the numbers $c_k(\tau,T)$ share one sign then the last sum equals $\left|\sum_k w_k c_k(\tau,T)\right| = |\mathcal{W}_{\tau,T}(\kappa)|$. The bound holds at every $\kappa \in (0,1)$ because the hypothesis does not involve κ. $\square$

Remark 3 (Verification of Condition 1)**.** *Condition 1 is checked, and the coefficients c_k of* (25) *are reported, at order size two on the calibration grid: the frozen baseline policy, the two windows $T \in \{5, 10\}$, the five threshold quantiles* 0.1, 0.3, 0.5, 0.7, 0.9 *of the terminal stake distribution under the benchmark policy, taken in adjacent pairs, and the* 71 *node liquidity grid $\kappa = 0.15, 0.16, \ldots, 0.85$. The record is* `numerical_v4/checks/t2_threshold_revelation_check.json`*. Its status on that grid is NUMERICAL.*

3 The who-gets-caught corollary of the clock dial

3.1 Who gets caught

The clock theorem makes window attenuation equivalent to $W_T C_T \leq 1$ and signs only the weight leg. This section signs the composition leg. It writes the longer clock's pooled sensitivity as a mass-weighted average of what the shorter clock leaves in the pool and what it takes out, and reads both $C_T \leq 1$ and $W_T C_T \leq 1$ off that average.

Setup. Fix the threshold margin τ and the plan and cutoff policies, and compare two window margins $T' < T$ at a common κ. Write

$$A \;=\; \mathcal{C}_P(\tau,T), \qquad B \;=\; \mathcal{C}_F(\tau,T') \setminus \mathcal{C}_F(\tau,T), \tag{30}$$

so A is the pooled cell under the longer clock and B collects the histories the shorter clock newly flags. Write

$$\varphi \;=\; \frac{\mathbb{P}(B)}{\mathbb{P}(A)} \tag{31}$$

for the share of the longer clock's pooled cell that the shorter clock catches.

Each rule pins its own premium kernel. The control-node information set carries the coordinate "the filing has landed by date d", so the pooled history at window margin T and the pooled history at window margin T' are different observables, and the engagement posterior $\pi = \mathbb{P}(a = 1 \mid \cdot)$ and the price are taken with respect to whichever of the two the rule supplies. Write $h^{(T)}$ and $h^{(T')}$ for the kernel $h = \pi p$ evaluated at the control-node information set of the rule (τ, T) and of the rule (τ, T'). The two pooled averages of the clock theorem are then

$$M_P(\tau,T) \;=\; \Delta_m\,\mathbb{E}\big[h^{(T)} \mid A\big], \qquad M_P(\tau,T') \;=\; \Delta_m\,\mathbb{E}\big[h^{(T')} \mid A \setminus B\big], \tag{32}$$

the second reading of the pooled cell at the shorter clock being part (i) below. Set

$$s_A \;=\; \partial_\kappa \mathbb{E}\big[h^{(T)} \mid A\big], \qquad s_{A\setminus B} \;=\; \partial_\kappa \mathbb{E}\big[h^{(T')} \mid A \setminus B\big], \tag{33}$$

so that, with $\Delta_m > 0$ finite,

$$\mathcal{S}_P(\tau,T) \;=\; \Delta_m\,|s_A|, \qquad \mathcal{S}_P(\tau,T') \;=\; \Delta_m\,|s_{A\setminus B}|, \qquad C_T \;=\; \frac{|s_{A\setminus B}|}{|s_A|}, \tag{34}$$

by the definitions of the two pooled sensitivities and no more.

Two further quantities separate what the catch removes from how the surviving pool is re-read:

$$\tilde{s}_B \;=\; \partial_\kappa \mathbb{E}\big[h^{(T)} \mid B\big], \qquad \delta \;=\; \partial_\kappa \mathbb{E}\big[h^{(T')} - h^{(T)} \mid A \setminus B\big]. \tag{35}$$

Here $\tilde{s}_B$ is the noise sensitivity of the newly caught histories, priced as the longer clock priced them, and δ is the *re-pricing remainder*: the pooled posterior conditions on the pool the rule leaves behind, so the histories that stay pooled are read against a smaller pool once the clock shortens, and δ is the rate at which that re-reading moves their noise sensitivity. Define the *caught leg* by

$$s_B \;:=\; \frac{1}{\mathbb{P}(B)}\,\partial_\kappa\Big(\mathbb{E}\big[h^{(T)}\mathbf{1}_A\big] - \mathbb{E}\big[h^{(T')}\mathbf{1}_{A\setminus B}\big]\Big). \tag{36}$$

The pooled cell contributes to the expected engagement premium an amount

$$\begin{aligned} \Delta_m\,\mathbb{E}\big[h^{(T)}\mathbf{1}_A\big] \;&=\; \big(1-\Omega(\tau,T)\big)\,M_P(\tau,T) \quad \text{at the longer clock,} \\ \Delta_m\,\mathbb{E}\big[h^{(T')}\mathbf{1}_{A\setminus B}\big] \;&=\; \big(1-\Omega(\tau,T')\big)\,M_P(\tau,T') \quad \text{at the shorter one.} \end{aligned}$$

So s_B is the derivative of $\Lambda_T - \Lambda_{T'}$ per unit of pooled mass removed: it is the noise sensitivity of the premium mass the shorter clock removes from the pool, and it carries the survivors' re-pricing. It is not $\tilde{s}_B$. Under the extra clause of (C-2) it splits into the two parts just named,

$$s_B \;=\; \tilde{s}_B \;-\; \frac{1-\varphi}{\varphi}\,\delta, \tag{37}$$

which part (iii) proves.

The hypotheses are:

(C-1) *Fixed policies.* The plan menu, the execution policies and the cutoff vector are held fixed in κ and held fixed across the two window margins, the threshold margin τ is common, the noise intensity κ is common, and the timing carries no feedback from order flow into the blockholder's executed order path. Liquidity enters the primitives in one place only, the law of the ternary noise mark, which is standing condition (S11) of the partition subsection; part (ii) consumes that.

(C-2) *Differentiability.* The two conditional expectations in (33) are differentiable in κ at the point compared. For the split (37) into $\tilde{s}_B$ and δ, $\mathbb{E}[h^{(T)} \mid B]$ is also differentiable. The kernel is the pinned version of standing condition (S8), so $h^{(T)}$ and $h^{(T')}$ are well-defined random variables rather than equivalence classes; part (iii) consumes that.

(C-3) *Non-degeneracy.* $\mathbb{P}(B) > 0$, $0 < \Omega(\tau,T)$, $\Omega(\tau,T') < 1$, and $s_A \neq 0$, equivalently $\mathcal{S}_P(\tau,T) > 0$. The clause $0 < \Omega(\tau,T)$ is there so that the clock theorem applies. Wherever this corollary is used to read the criterion $W_T C_T \leq 1$ in Theorem 1, that theorem's hypotheses also apply. They include the hypotheses of Proposition 1 at both clocks, including the invariance of the flagged endpoint in κ at fixed policies.

Label: PROVED.

Corollary 1 (Who gets caught)**.** *Under (C-1) to (C-3):*

(i) The cut is nested. *$\mathcal{C}_F(\tau,T) \subseteq \mathcal{C}_F(\tau,T')$, hence $B \subseteq A$ and $A \setminus B = \mathcal{C}_P(\tau,T')$, and the weight leg is*

$$W_T \;=\; 1-\varphi \;\le\; 1, \qquad \varphi \in (0,1). \tag{38}$$

(ii) The cell masses are κ-free. *$\partial_\kappa \mathbb{P}(A) = \partial_\kappa \mathbb{P}(B) = 0$, so φ does not depend on κ.*

(iii) The cut identity.

$$s_{A\setminus B} \;=\; \frac{\mathbb{P}(A)\,s_A - \mathbb{P}(B)\,s_B}{\mathbb{P}(A)-\mathbb{P}(B)} \;=\; \frac{s_A - \varphi\,s_B}{1-\varphi} \;=\; \frac{s_A - \varphi\,\tilde{s}_B}{1-\varphi} \;+\; \delta, \tag{39}$$

the last expression under the extra differentiability of (C-2). Equivalently $s_A = (1-\varphi)\,s_{A\setminus B} + \varphi\,s_B$: the longer clock's pooled sensitivity is the mass-weighted average of the sensitivity of what the shorter clock leaves in the pool and s_B, the derivative of $\Lambda_T - \Lambda_{T'}$ per unit of pooled mass removed, which carries the survivors' re-pricing.

(iv) The composition ratio.

$$\begin{aligned} C_T \;&=\; \frac{|s_A - \varphi s_B|}{(1-\varphi)\,|s_A|}, \\ C_T \le 1 \quad&\Longleftrightarrow\quad (s_B - s_A)\big(\varphi s_B - (2-\varphi)s_A\big) \le 0, \end{aligned} \tag{40}$$

that is, if and only if s_B lies weakly between s_A and $\big((2-\varphi)/\varphi\big)s_A$.

(v) The attenuation criterion.

$$\begin{aligned} W_T C_T \;&=\; \frac{|s_A - \varphi s_B|}{|s_A|}, \\ W_T C_T \le 1 \quad&\Longleftrightarrow\quad s_B\big(2s_A - \varphi s_B\big) \ge 0, \end{aligned} \tag{41}$$

that is, if and only if s_B lies weakly between 0 and $(2/\varphi)s_A$.

(vi) Readings. *Write $\rho = s_B/s_A$, which is well defined by (C-3).*

- *(a) If $s_A s_B \le 0$, including $s_B = 0$, then $C_T > 1$ strictly.*
- *(b) If $s_A s_B > 0$, then $C_T \le 1$ if and only if $|s_A| \le |s_B| \le \big((2-\varphi)/\varphi\big)|s_A|$.*
- *(c) If the cut does not reverse the pooled sensitivity, $s_{A\setminus B}\, s_A \ge 0$, then $C_T \le 1$ if and only if $\rho \ge 1$. The inequality $\rho \ge 1 > 0$ already forces a common sign, so that reading is $|s_B| \ge |s_A|$. Under a common sign the proviso $\sigma \ge 0$ is $|s_B| \le |s_A|/\varphi$.*
- *(d) $W_T C_T \le 1$ if and only if s_B shares the sign of s_A or is zero, and $|s_B| \le (2/\varphi)|s_A|$.*
- *(e) The two upper limits $(2-\varphi)/\varphi$ and $2/\varphi$ are strictly decreasing in φ and diverge as $\varphi \downarrow 0$. For every $M \ge 1$ and every cut with $\varphi \le 2/(1+M)$ and $|s_B| \le M|s_A|$, the upper limits in (b) and (d) are slack, so there $C_T \le 1$ reads: a common sign together with $|s_B| \ge |s_A|$; and $W_T C_T \le 1$ reads: a common sign or $s_B = 0$.*

Proof. Part (i). The disclosure indicator at window margin T is $D(\tau,T) = \mathbf{1}\{a = 1,\ c(\tau) + T \le H\}$, where $c(\tau)$ is the round in which the executed stake path crosses τ. If $D(\tau,T) = 1$

then $c(\tau) + T' \le c(\tau) + T \le H$ because $T' < T$, so $D(\tau, T') = 1$. Hence $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau, T')$ and, taking complements, $\mathcal{C}_P(\tau, T') \subseteq \mathcal{C}_P(\tau, T) = A$. Therefore $B = \mathcal{C}_F(\tau, T') \cap A \subseteq A$ and $A \setminus B = A \cap \mathcal{C}_P(\tau, T') = \mathcal{C}_P(\tau, T')$. Taking probabilities, $\Omega(\tau, T) \le \Omega(\tau, T') < 1$, so both pooled cells carry positive mass; since $B \subseteq A$, $\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(B)$, and

$$W_T \;=\; \frac{1 - \Omega(\tau, T')}{1 - \Omega(\tau, T)} \;=\; \frac{\mathbb{P}(A \setminus B)}{\mathbb{P}(A)} \;=\; 1 - \varphi.$$

By (C-3), $\mathbb{P}(B) > 0$ gives $\varphi > 0$ and $\Omega(\tau, T') < 1$ gives $\mathbb{P}(A \setminus B) > 0$, hence $\varphi < 1$.

Part (ii). Under fixed policies the selected plan and the executed order path are functions of the plan index and the signal alone, and by the no-feedback clause of (C-1) the order placed in a round does not read realised order flow. So the crossing round $c(\tau)$, and with it $D(\tau, T)$ and $D(\tau, T')$, are measurable with respect to the plan and signal. The intensity κ parametrises the noise law only; the laws of the value, the signal noise and the bidder draw carry no κ, and (C-1) holds the cutoff vector fixed in κ, so the joint law of the plan and the signal is one and the same at every κ. Hence $\mathbb{P}(A)$ and $\mathbb{P}(B)$ do not vary with κ, and neither does their ratio φ. This is the same constancy the clock theorem uses for the flagged weight, applied at both window margins.

Part (iii). By part (i), B and $A \setminus B$ partition A and both have positive mass. Write the pooled contributions to the expected premium at the two clocks, in kernel units, as

$$\begin{aligned}
\Lambda_T \;&=\; \mathbb{E}\big[h^{(T)} \mathbf{1}_A\big] \;=\; \mathbb{P}(A)\, \mathbb{E}\big[h^{(T)} \mid A\big], \\
\Lambda_{T'} \;&=\; \mathbb{E}\big[h^{(T')} \mathbf{1}_{A \setminus B}\big] \;=\; \mathbb{P}(A \setminus B)\, \mathbb{E}\big[h^{(T')} \mid A \setminus B\big].
\end{aligned}$$

By part (ii) the two masses are constants in κ, so the derivative that (C-2) supplies passes through them and leaves them alone:

$$\partial_\kappa \Lambda_T \;=\; \mathbb{P}(A)\, s_A, \qquad \partial_\kappa \Lambda_{T'} \;=\; \big(\mathbb{P}(A) - \mathbb{P}(B)\big)\, s_{A \setminus B}.$$

Definition (36) reads $s_B = \partial_\kappa(\Lambda_T - \Lambda_{T'})/\mathbb{P}(B)$, a division by a positive number, so

$$\mathbb{P}(B)\, s_B \;=\; \mathbb{P}(A)\, s_A - \big(\mathbb{P}(A) - \mathbb{P}(B)\big)\, s_{A \setminus B},$$

and solving for $s_{A \setminus B}$, which the positive mass $\mathbb{P}(A) - \mathbb{P}(B)$ permits, gives the first equality of (39). Dividing numerator and denominator by $\mathbb{P}(A)$ gives the second. Multiplying through by $1 - \varphi$ and collecting terms gives $s_A = (1 - \varphi) s_{A \setminus B} + \varphi s_B$.

For the third expression, add and subtract $\mathbb{E}[h^{(T)} \mathbf{1}_{A \setminus B}]$ inside $\Lambda_T - \Lambda_{T'}$. Since $\mathbf{1}_A = \mathbf{1}_{A \setminus B} + \mathbf{1}_B$ by part (i),

$$\begin{aligned}
\Lambda_T - \Lambda_{T'} \;&=\; \mathbb{E}\big[h^{(T)} \mathbf{1}_B\big] \;-\; \mathbb{E}\big[\big(h^{(T')} - h^{(T)}\big) \mathbf{1}_{A \setminus B}\big] \\
&=\; \mathbb{P}(B)\, \mathbb{E}\big[h^{(T)} \mid B\big] \;-\; \mathbb{P}(A \setminus B)\, \mathbb{E}\big[h^{(T')} - h^{(T)} \mid A \setminus B\big].
\end{aligned}$$

Both masses are κ-free by part (ii), and the extra clause of (C-2) makes the first term differentiable, hence so is the second. Differentiating and dividing by $\mathbb{P}(B)$ gives $s_B = \tilde{s}_B - \big((1 - \varphi)/\varphi\big)\delta$, which is (37). Substituting it into the second equality of (39) gives the third.

Part (iv). By (34) and part (i), $C_T = |s_{A\setminus B}|/|s_A|$, and by (39) with $1-\varphi > 0$,

$$C_T \;=\; \frac{|s_A - \varphi s_B|}{(1-\varphi)|s_A|}.$$

The denominator is non-zero because $s_A \neq 0$ in (C-3). Both sides of $C_T \leq 1$ are non-negative, so $C_T \leq 1$ if and only if $s_{A\setminus B}^2 \leq s_A^2$, and multiplying by $(1-\varphi)^2 > 0$, if and only if $u^2 - w^2 \leq 0$, where $u = s_A - \varphi s_B$ and $w = (1-\varphi)s_A$. Since $u - w = \varphi(s_A - s_B)$ and $u + w = (2-\varphi)s_A - \varphi s_B$,

$$u^2 - w^2 \;=\; (u-w)(u+w) \;=\; \varphi\,(s_A - s_B)\big((2-\varphi)s_A - \varphi s_B\big).$$

Since $\varphi > 0$, the sign of the whole is the sign of $(s_A - s_B)\big((2-\varphi)s_A - \varphi s_B\big)$, and negating both factors leaves the product unchanged, which gives the stated form $(s_B - s_A)\big(\varphi s_B - (2-\varphi)s_A\big) \leq 0$. Writing that product as $\varphi(s_B - s_A)\big(s_B - \big((2-\varphi)/\varphi\big)s_A\big)$ shows it is non-positive exactly when s_B lies weakly between the two roots s_A and $\big((2-\varphi)/\varphi\big)s_A$.

Part (v). By part (i) and part (iv),

$$W_T C_T \;=\; (1-\varphi)\,\frac{|s_{A\setminus B}|}{|s_A|} \;=\; \frac{\big|(1-\varphi)s_{A\setminus B}\big|}{|s_A|} \;=\; \frac{|s_A - \varphi s_B|}{|s_A|}.$$

As in part (iv), $W_T C_T \leq 1$ if and only if $u^2 - s_A^2 \leq 0$ for $u = s_A - \varphi s_B$, and since $u - s_A = -\varphi s_B$ and $u + s_A = 2s_A - \varphi s_B$,

$$u^2 - s_A^2 \;=\; (u - s_A)(u + s_A) \;=\; -\varphi\, s_B\big(2s_A - \varphi s_B\big),$$

so the criterion is $s_B(2s_A - \varphi s_B) \geq 0$. Writing it as $\varphi\, s_B\big((2/\varphi)s_A - s_B\big) \geq 0$ shows it holds exactly when s_B lies weakly between 0 and $(2/\varphi)s_A$.

Part (vi). Since $\varphi \in (0,1)$, the factor $(2-\varphi)/\varphi$ is strictly positive, so the two roots in part (iv) are non-zero and share the sign of s_A, and the closed interval they bound does not contain zero. If $s_A s_B \leq 0$ then s_B is zero or lies on the far side of zero from both roots, so it is strictly outside that interval, the product in (40) is strictly positive, and $C_T > 1$. That is (a). For (b), divide the interval condition by s_A: with a common sign, $\rho > 0$ and s_B between s_A and $\big((2-\varphi)/\varphi\big)s_A$ reads $1 \leq \rho \leq (2-\varphi)/\varphi$, and multiplying by $|s_A|$ gives the stated magnitudes. For (c), set $\sigma = s_{A\setminus B}/s_A = (1-\varphi\rho)/(1-\varphi)$ by (39). The proviso $s_{A\setminus B}s_A \geq 0$ is $\sigma \geq 0$, so $C_T = |\sigma| = \sigma$, and $C_T \leq 1$ if and only if $1 - \varphi\rho \leq 1 - \varphi$, that is $\rho \geq 1$. The inequality $\rho \geq 1 > 0$ already forces a common sign, under which $\rho \geq 1$ is $|s_B| \geq |s_A|$. Under a common sign the proviso $\sigma \geq 0$ is $\varphi\rho \leq 1$, that is $|s_B| \leq |s_A|/\varphi$. For (d), divide the interval condition of part (v) by s_A: it reads $0 \leq \rho \leq 2/\varphi$, which is a common sign or $s_B = 0$, together with $|s_B| \leq (2/\varphi)|s_A|$. For (e), both $(2-\varphi)/\varphi$ and $2/\varphi$ are strictly decreasing on $(0,1)$ and tend to $+\infty$ as $\varphi \downarrow 0$. Fix $M \geq 1$ and $\varphi \leq 2/(1+M)$. Then $(2-\varphi)/\varphi \geq M$, because that inequality is $2 - \varphi \geq M\varphi$, and $2/\varphi \geq (2-\varphi)/\varphi \geq M$. So $|s_B| \leq M|s_A|$ makes the upper limits in (b) and (d) slack, leaving in (b) the common sign with $|s_B| \geq |s_A|$ and in (d) the common sign or $s_B = 0$. $\square$

Reading. Part (iii) is the whole content in one line: the longer clock's pooled sensitivity is an average over what the shorter clock leaves in the pool and what it takes out of the pool, with weights the two masses, and those weights do not move with liquidity. So shortening the clock pulls the pooled sensitivity down only if what it takes out sat above the average. Parts (iv) and (vi) put the bound on how far above. What the clock removes must move with the pool in κ and be at least as noise-sensitive as the pool, and it must not be so much more sensitive that the remaining pool is driven through zero to a larger magnitude than the pool started with. A cut that takes a small share of the pool faces only the first two requirements when, as part (vi)(e) states, there is some $M \geq 1$ with $\varphi \leq 2/(1+M)$ and $|s_B| \leq M|s_A|$. Part (v) does the same for the criterion the clock theorem states, $W_T C_T \leq 1$, where the weight leg is already working in the same direction: there what the clock removes need only move with the pool and stay inside $2/\varphi$ times its sensitivity.

What the clock removes has two parts, and (37) separates them. The first is the noise sensitivity of the newly caught histories themselves, read at the prices the longer clock gave them. The second is the re-pricing remainder: the surviving pool is a smaller pool once the clock shortens, and the market reads it against that smaller pool, so its own sensitivity moves too. The criterion runs on the sum, which is the sensitivity of the premium mass that leaves the pool. When the re-pricing remainder is zero the sum is the first part alone, and the criterion is a statement about the caught histories and nothing else.

The corollary turns the ratio C_T into a question with an answer at every calibration node, and that question is who gets caught. Under a common sign, $s_A s_B > 0$, the composition ratio satisfies $C_T \leq 1$ if and only if the caught sensitivity s_B lies weakly between s_A and $\big((2-\varphi)/\varphi\big)\, s_A$, which is part (iv); and the shorter clock lowers noise sensitivity overall, $W_T C_T \leq 1$, if and only if s_B lies weakly between 0 and $(2/\varphi)\, s_A$, which is part (v). The two bands differ: the weight leg lets the overall criterion accept a caught leg less sensitive than the pool, which the composition criterion alone does not.

4 Pricing root and timing lemmas

This section states and proves two lemmas the main text uses: the competitive pricing root is unique and continuous, and at fixed policies a shorter window weakly lowers the stake at filing on every path flagged under both windows.
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Lemma 8 (Uniqueness and continuity of the competitive pricing root)**.** *Assume the standing conditions (S1) and (S8); the latter gives an entry probability in $(0,1)$ that is continuous in the posterior and the price, and this lemma adds its normal form and its monotonicity in the price. Let $m_0 \geq 0$, $\Delta_m > 0$, $\sigma_\xi > 0$, $\Delta_V \geq 0$, and let the potential bidder's entry probability be*

$$p(P,\pi) \;=\; 1 - \Phi\!\left(\frac{P + K + m_0 + \pi\Delta_m - \bar{S}}{\sigma_\xi}\right), \tag{42}$$

where Φ is the standard normal cumulative distribution function and $K, \bar{S}$ are finite constants. Then, for every $(\hat{v},\pi) \in \mathbb{R}\times[0,1]$, the competitive control-node pricing equation

$$P \;=\; \big(1 - p(P,\pi)\big)\big(\hat{v} + \pi\Delta_V\big) \;+\; p(P,\pi)\big(P + m_0 + \pi\Delta_m\big) \tag{43}$$

has a unique finite solution $P = \Pi(\hat{v},\pi)$. The solution map $\Pi : \mathbb{R}\times[0,1] \to \mathbb{R}$ is continuous, strictly *increasing in $\hat{v}$, and satisfies the linear-growth bound*

$$|\Pi(\hat{v},\pi)| \;\le\; |\hat{v}| + C_\Pi \tag{44}$$

for the finite constant $C_\Pi := \max_{\pi\in[0,1]} |\Pi(0,\pi)|$.

Proof. Set $\hat{V} = \hat{v} + \pi\Delta_V$ and $\widetilde{m} = m_0 + \pi\Delta_m$. Since $m_0 \ge 0$, $\Delta_m > 0$, and $\pi \in [0,1]$, we have $\widetilde{m} \ge m_0 \ge 0$. Because Φ is strictly between 0 and 1 everywhere on the real line, $p(P,\pi) \in (0,1)$ for every finite P. Rearranging (43) by collecting terms in P gives

$$\big(1-p(P,\pi)\big)P \;=\; \big(1-p(P,\pi)\big)\hat{V} \;+\; p(P,\pi)\,\widetilde{m},$$

or equivalently, dividing by $1-p(P,\pi)>0$,

$$R(P;\hat{V},\pi) \;:=\; P \;-\; \hat{V} \;-\; \widetilde{m}\,\frac{p(P,\pi)}{1-p(P,\pi)} \;=\; 0. \tag{45}$$

The entry probability $p(P,\pi)$ in (42) is strictly decreasing and continuously differentiable in P, with $\lim_{P\to-\infty} p(P,\pi) = 1$ and $\lim_{P\to+\infty} p(P,\pi) = 0$. The odds ratio $q \mapsto q/(1-q)$ is strictly increasing on $(0,1)$. Therefore, $P \mapsto p(P,\pi)/(1-p(P,\pi))$ is strictly decreasing in P. Since $\widetilde{m} \ge 0$, the function

$$P \;\longmapsto\; -\,\widetilde{m}\,\frac{p(P,\pi)}{1-p(P,\pi)}$$

is weakly increasing in P. Consequently, $R(\cdot;\hat{V},\pi)$ is strictly increasing on $\mathbb{R}$.

Moreover, as $P\to-\infty$, $p(P,\pi)\to 1$ and $p/(1-p)\to+\infty$, so $R(P;\hat{V},\pi)\to-\infty$. As $P\to+\infty$, $p(P,\pi)\to 0$ and $p/(1-p)\to 0$, so $R(P;\hat{V},\pi)\to+\infty$. By the intermediate value theorem, a solution to $R(P;\hat{V},\pi)=0$ exists. By strict monotonicity of R in P, the solution is unique.

To establish continuity, differentiate R with respect to P:

$$\frac{\partial R}{\partial P} \;=\; 1 \;-\; \widetilde{m}\,\frac{\partial}{\partial P}\left(\frac{p}{1-p}\right) \;\ge\; 1 \;>\; 0,$$

since $\widetilde{m}\ge 0$ and $\partial_P(p/(1-p)) < 0$. By the implicit function theorem, the unique root $\Pi(\hat{v},\pi)$ is continuously differentiable in $(\hat{V},\pi)$, and therefore continuous in $(\hat{v},\pi)$ on $\mathbb{R}\times[0,1]$. Furthermore,

$$\frac{\partial \Pi}{\partial \hat{V}} \;=\; \frac{1}{\partial R/\partial P} \;\in\; (0,1],$$

so Π is strictly increasing in $\hat{v}$. Write $\Pi^*(\hat{V},\pi)$ for the root as a function of $\hat{V}$, so that $\Pi(\hat{v},\pi) = \Pi^*(\hat{v}+\pi\Delta_V,\pi)$. For fixed π, the derivative bound above makes Π^* 1-Lipschitz in $\hat{V}$, and the two arguments $\hat{v}+\pi\Delta_V$ and $\pi\Delta_V$ differ by $\hat{v}$, so $|\Pi(\hat{v},\pi)-\Pi(0,\pi)| \le |\hat{v}|$. Since $[0,1]$ is compact and $\pi\mapsto\Pi(0,\pi)$ is continuous, $C_\Pi := \max_{\pi\in[0,1]}|\Pi(0,\pi)| < \infty$. The bound (44) follows. $\square$

Label: PROVED.

Lemma 9 (Window monotonicity of the stake at filing under fixed policies)**.** *Assume the standing conditions (S2), (S4), (S5), (S7), and (S11). Let the plan menu, the cutoff vector, and the execution paths $(B_j(s,\cdot))_{j\in\mathcal{J}}$ be fixed. Compare two window margins $T' < T$ at a common threshold*

τ. Write $c(s;\tau)$ for the crossing date of a Voice type, $f(s;T_0) = c(s;\tau) + T_0$ for the filing date under window T_0, and $B^F(s;T_0) = B_{\text{Voice}}(s, f(s;T_0))$ for the stake at filing, defined on the flagged cell $\{f(s;T_0) \le H\}$ and nowhere else. Then, for every signal realization s:

(a) On Exit and Hold plans no filing lands under either window margin, and the stake path is the same under both.

(b) On Voice plans, the crossing date $c(s;\tau)$ is the same under both window margins, and the flagged cells are nested: $\{f(s;T) \le H\} \subseteq \{f(s;T') \le H\}$. On the difference $\{c(s;\tau) + T' \le H < c(s;\tau) + T\}$ the filing lands under T' and does not land under T, so $B^F(s;T')$ is defined and $B^F(s;T)$ is not.

(c) On Voice plans with $c(s;\tau) + T \le H$, the paths flagged under both windows, the filing lands weakly earlier under T', $f(s;T') < f(s;T) \le H$, and

$$B^F(s;T') \;\le\; B^F(s;T). \tag{46}$$

In particular, shortening the window weakly lowers the stake at filing on every path flagged under both windows, and weakly enlarges the set of paths on which a filing lands.

Proof. Under fixed policies ((S11)) the plan selection and the stake paths are identical under T and T'. By (S5) only Voice plans cross τ, so on Exit and Hold plans no filing lands under either window and the stake path is common; this is part (a).

For a Voice type, (S5) and (S7) make the crossing date $c(s;\tau) = \min\{d \le H : B_{\text{Voice}}(s,d) \ge \tau\}$ (or $+\infty$) a function of the signal and τ alone, so it is the same under both windows. By (S5) the filing lands exactly at $f(s;T_0) = c(s;\tau) + T_0$, and the flagged cell is the event $f(s;T_0) \le H$. Since $T' < T$, $f(s;T') < f(s;T)$ whenever $c(s;\tau) < \infty$, so $f(s;T) \le H$ implies $f(s;T') \le H$: the flagged cells are nested. On $\{c + T' \le H < c + T\}$ the filing lands under T' and not under T, which is the last clause of part (b).

For part (c), let $c(s;\tau) + T \le H$. Then both filing dates lie in the calendar, $f(s;T') < f(s;T) \le H$, and both stakes at filing are defined. By (S4) the Voice stake path $d \mapsto B_{\text{Voice}}(s,d)$ is weakly increasing in the calendar date, so $B^F(s;T') = B_{\text{Voice}}(s, f(s;T')) \le B_{\text{Voice}}(s, f(s;T)) = B^F(s;T)$, which is (46). $\square$